ANALYSIS OF COMPATIBLE DISCRETE OPERATOR SCHEMES ON POLYHEDRAL MESHES FOR MAXWELL'S EQUATIONS IN THE TIME DOMAIN

VALENTIN RITZENTHALER†

Abstract. We analyze a Compatible Discrete Operator (CDO) scheme for Maxwell's equations in the time domain on polyhedral meshes, with piecewise constant anisotropic material tensors and a perfectly conducting boundary. The topological relations (Faraday's and Ampère's laws) are discretized exactly, by the incidence matrices of a primal and a dual mesh in space and by a leapfrog integration in time, whereas the constitutive relations are discretized by discrete Hodge operators carrying the metric and material information. The analysis only relies on three algebraic requirements on the local Hodge operators: symmetry, uniform stability and exactness on constant fields. We prove that the discrete divergence constraints and the boundary magnetic flux are preserved exactly, that a discrete pseudo-energy is conserved for any time step, and that it controls the discrete fields under an explicit CFL condition. The consistency error is split into spatial residuals of the Hodge operators and temporal quadrature residuals of the constitutive relations, yielding an error estimate of order $h + \Delta t^2$ in discrete energy norms.

1. Introduction

In the numerical simulation of Maxwell's equations in the time domain, one of the main objectives is to reproduce the physics of electromagnetic fields with high accuracy and a low computational cost. The scheme introduced by Yee [45] is still the workhorse of the field: on a Cartesian grid, the electric and magnetic unknowns are staggered in space and in time, the discrete divergence constraints are preserved exactly and a discrete energy is conserved. Its convergence on nonuniform Cartesian grids was analyzed by Monk and Süli [33]. The price to pay is geometric: curved objects are approximated by staircases, which degrades accuracy in their vicinity. This motivates the extension of the Yee scheme to tetrahedral, polyhedral and hybrid meshes, which fit the geometry much better, see for instance [12, 13, 37]. Such extensions must preserve what makes the Yee scheme robust: the stability of explicit time-domain schemes for Maxwell's equations relies on the adjoint character of the two discrete curl operators, and the structural properties (divergence and energy conservation) rely on the exactness of the discrete differential operators.

The laws of electromagnetism have a structure which is independent of the differential formalism used to write them. As noticed early by Tonti [42, 43] and Bossavit [10], the physical variables are naturally attached to geometric entities: electromotive and magnetomotive forces are circulations along edges, electric and magnetic fluxes are fluxes through faces, charges are densities in volumes; in time, the fluxes are attached to instants and the circulations to intervals. With this classification, the laws split into two parts. The *topological* relations (Faraday, Ampère, Gauss, Thomson) only depend on the orientation of the geometric entities and on their boundaries; they are exactly written as balance equations with integer incidence coefficients. The *constitutive* relations, which depend on the materials and, mathematically, on the metric, establish a *duality* between variables attached to geometric entities of complementary dimension: this is why they are naturally discretized on a pair of meshes, a primal and a dual one, with the electric circulations on primal edges and the electric fluxes on dual faces, and conversely for the magnetic variables. This point of view was developed in computational electromagnetism under several names: generalized finite differences [11], the Finite Integration Technique [18], the cell method [41, 42], the Discrete Geometric Approach [20, 21], covolume methods [30, 35], mimetic finite differences [4, 5, 31], discrete exterior calculus [23, 38] and compatible discretizations [27, 40]. These approaches, together with Whitney forms and edge elements [9, 34] and finite element exterior calculus [2, 17, 29], share a common feature: the topological relations are discretized exactly by the incidence matrices of the meshes, and all the approximation is concentrated in the discrete counterparts of the constitutive relations, the *discrete Hodge operators* [28, 39].

Bonelle and Ern [7, 8] synthesized this geometric point of view into the *Compatible Discrete Operator* (CDO) framework, in which the discrete Hodge operators are assembled cellwise, on general polyhedral meshes, from

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† Independent researcher.

local operators satisfying three algebraic requirements (symmetry, uniform stability and exactness on constant fields); see [6, 19, 20] for the construction of such operators. The key feature of this framework is that the *consistency error is entirely carried by the Hodge operators*: the reduction maps commute with the discrete differential operators, so that the only non-commuting diagrams are the constitutive ones. This framework has provided a complete analysis of elliptic and Stokes problems [7, 8], advection-diffusion problems [14, 15] and discrete Hodge decompositions [32]. For Maxwell's equations in the time domain, the convergence of related polyhedral schemes has been studied in the Discrete Geometric Approach by Codecasa and Trevisan [21, 22] and, for covolume schemes on Delaunay–Voronoi meshes, by Nicolaides and Wang [35]; energy techniques for leapfrog time integration have been developed for discontinuous Galerkin methods in [26], and a Yee-like finite element scheme on unstructured grids has recently been analyzed in [36]. The present paper fills a gap between these works: it provides, within the CDO framework and under the sole algebraic requirements on the Hodge operators recalled above, a complete analysis of the leapfrog CDO scheme for Maxwell's equations on polyhedral meshes with anisotropic, piecewise constant material tensors.

To formulate the scheme, time is treated in the same spirit as space. A primal and a dual time mesh are introduced, and the time integration of Faraday's and Ampère's laws is exact on the fluxes, in the same way as the spatial discretization is exact for the topological relations. Consequently, the discrete unknowns approximating $\boldsymbol{E}$ and $\boldsymbol{H}$ are circulations integrated over time intervals, whereas the unknowns approximating $\boldsymbol{D}$ and $\boldsymbol{B}$ are fluxes at time instants, and the only time averages are carried by the constitutive relations. This one-dimensional instance, in time, of the primal/dual structure yields the classical leapfrog scheme of [11, 18, 21, 27]. The discrete divergence constraints and the vanishing normal magnetic flux on the boundary are preserved exactly along the discrete trajectory.

The stability analysis relies on a discrete pseudo-energy, which is exactly conserved by the scheme for any time step. We show that under a CFL condition this pseudo-energy is coercive and controls the discrete fields. For the convergence analysis, the consistency error is decomposed into spatial residuals, which only involve the Hodge operators applied to the reductions of the exact fields, and temporal residuals, which are quadrature errors on the time averages of the exact circulations. The spatial residuals are bounded cellwise, using the exactness on constant fields, for exact fields that are piecewise H^1 on a material partition, which allows jumps across material interfaces. Combining these bounds with the energy analysis, we establish an energy-norm error estimate of order $h + \Delta t^2$ for the fluxes and for the full-interval time averages of the circulations. Finally, under additional assumptions on reconstruction operators, a corollary also yields $\boldsymbol{L}^2(\Omega)$ convergence of reconstructed fields at their respective time levels.

Section 2 introduces the model problem, with anisotropic material tensors and a perfectly electrically conducting boundary, and recalls its structural properties: divergence and energy conservation. Section 3 describes the discrete setting: the primal and dual meshes in space and in time, the degrees of freedom and the reduction maps, the discrete differential operators with their exactness and commuting properties, and the discrete Hodge operators with the algebraic requirements they satisfy. Section 4 presents the CDO scheme and the exact conservation of the discrete divergences. Section 5 develops the analysis: stability through the pseudo-energy and the CFL condition, consistency through the spatial and temporal residuals, and convergence. The proofs of the stability, consistency and convergence results are collected in Appendices A to C. Section 6 draws conclusions and perspectives.

2. Model problem and structural properties

2.1. **Functional setting.** Let $\Omega \subset \mathbb{R}^3$ be an open bounded connected polyhedral set with Lipschitz boundary $\partial\Omega$ and outward unit normal $\boldsymbol{n}$. Let $t_F > 0$ be a final time and consider the time interval $[0, t_F]$. The material tensors $\boldsymbol{\varepsilon}, \boldsymbol{\mu} \in [L^\infty(\Omega)]^{3\times 3}$ (permittivity and permeability) are independent of time, symmetric and uniformly positive definite: there exist constants $0 < \varepsilon_\flat \le \varepsilon_\sharp < \infty$ and $0 < \mu_\flat \le \mu_\sharp < \infty$ such that

$$\varepsilon_\flat|\xi|^2 \le \xi^{\mathsf{T}}\boldsymbol{\varepsilon}(x)\xi \le \varepsilon_\sharp|\xi|^2 \quad\text{and}\quad \mu_\flat|\xi|^2 \le \xi^{\mathsf{T}}\boldsymbol{\mu}(x)\xi \le \mu_\sharp|\xi|^2, \quad \text{for a.e. } x \in \Omega,\ \forall \xi \in \mathbb{R}^3. \tag{2.1}$$

By (2.1), $\boldsymbol{\varepsilon}^{-1}$ and $\boldsymbol{\mu}^{-1}$ exist, are symmetric, belong to $[L^\infty(\Omega)]^{3\times 3}$, and satisfy the same type of bounds with constants $\varepsilon_\sharp^{-1} \le \varepsilon_\flat^{-1}$ and $\mu_\sharp^{-1} \le \mu_\flat^{-1}$.

We write $\boldsymbol{L}^2(\Omega) := [L^2(\Omega)]^3$ and use the spaces

$$\begin{aligned} H(\nabla\times;\Omega) &:= \{\boldsymbol{u} \in \boldsymbol{L}^2(\Omega) : \nabla \times \boldsymbol{u} \in \boldsymbol{L}^2(\Omega)\},\\ H_0(\nabla\times;\Omega) &:= \{\boldsymbol{u} \in H(\nabla\times;\Omega) : \boldsymbol{n} \times \boldsymbol{u} = 0 \text{ on } \partial\Omega\},\\ H(\nabla\cdot;\Omega) &:= \{\boldsymbol{u} \in \boldsymbol{L}^2(\Omega) : \nabla \cdot \boldsymbol{u} \in L^2(\Omega)\}, \end{aligned}$$

equipped with their graph norms, where $\nabla\times$ and $\nabla\cdot$ are understood in the sense of distributions, and the tangential trace $\boldsymbol{n} \times \boldsymbol{u}$ (resp. the normal trace $\boldsymbol{u} \cdot \boldsymbol{n}$) is the one defined on $H(\nabla\times;\Omega)$ (resp. on $H(\nabla\cdot;\Omega)$).

Equivalently, $H_0(\nabla\times;\Omega)$ is the closure of $[C_c^\infty(\Omega)]^3$ in $H(\nabla\times;\Omega)$, so that we have the Green formula:

$$\int_\Omega \nabla\times\boldsymbol{u}\cdot\boldsymbol{v} = \int_\Omega \boldsymbol{u}\cdot\nabla\times\boldsymbol{v}, \qquad \forall\boldsymbol{u}\in H_0(\nabla\times;\Omega),\ \forall\boldsymbol{v}\in H(\nabla\times;\Omega). \tag{2.2}$$

2.2. **Maxwell's equations.** Let $\boldsymbol{B},\boldsymbol{E},\boldsymbol{H},\boldsymbol{D} : \Omega\times[0,t_F]\to\mathbb{R}^3$ be the magnetic induction, the electric field, the magnetic field and the electric displacement, respectively. We consider the model problem: find $\boldsymbol{B},\boldsymbol{E},\boldsymbol{H},\boldsymbol{D} : \Omega\times[0,t_F]\to\mathbb{R}^3$ such that

$$\frac{\partial\boldsymbol{B}}{\partial t} = -\nabla\times\boldsymbol{E}, \tag{2.3a}$$

$$\frac{\partial\boldsymbol{D}}{\partial t} = \nabla\times\boldsymbol{H}, \tag{2.3b}$$

$$\boldsymbol{D} = \boldsymbol{\varepsilon}\boldsymbol{E}, \tag{2.3c}$$

$$\boldsymbol{B} = \boldsymbol{\mu}\boldsymbol{H}, \tag{2.3d}$$

in $\Omega\times[0,t_F]$, with the perfect electric conductor (PEC) boundary condition

$$\boldsymbol{n}\times\boldsymbol{E} = 0 \text{ on } \partial\Omega\times[0,t_F], \tag{2.4}$$

and initial conditions

$$\boldsymbol{D}(\cdot,t=0) = \boldsymbol{D}_0 \ \text{ and } \ \boldsymbol{B}(\cdot,t=0) = \boldsymbol{B}_0, \tag{2.5}$$

where $\boldsymbol{D}_0,\boldsymbol{B}_0\in H(\nabla\cdot;\Omega)$ are two given fields. By (2.3c) and (2.3d), the initial fields are $\boldsymbol{E}(\cdot,0)=\boldsymbol{\varepsilon}^{-1}\boldsymbol{D}_0$ and $\boldsymbol{H}(\cdot,0)=\boldsymbol{\mu}^{-1}\boldsymbol{B}_0$. Additionally, we make the following assumptions on the data. Let $\rho\in L^2(\Omega)$ be the volume charge density inside Ω. Since there is no current, it does not depend on time. The initial fields satisfy

$$\nabla\cdot\boldsymbol{D}_0 = \rho \ \text{ and } \ \nabla\cdot\boldsymbol{B}_0 = 0 \text{ in } \Omega, \tag{2.6}$$

the condition

$$\boldsymbol{B}_0\cdot\boldsymbol{n} = 0 \text{ on } \partial\Omega, \tag{2.7}$$

and the compatibility condition

$$\boldsymbol{\varepsilon}^{-1}\boldsymbol{D}_0\in H_0(\nabla\times;\Omega) \ \text{ and } \ \boldsymbol{\mu}^{-1}\boldsymbol{B}_0\in H(\nabla\times;\Omega). \tag{2.8}$$

Condition (2.8) states that the initial electric field already satisfies the boundary condition (2.4), and that the right-hand sides of (2.3a)–(2.3b) make sense at $t=0$. Under (2.1) and (2.8), problem (2.3)–(2.5) admits a unique solution with

$$\boldsymbol{E}\in C^0([0,t_F];H_0(\nabla\times;\Omega))\cap C^1([0,t_F];\boldsymbol{L}^2(\Omega)),$$
$$\boldsymbol{H}\in C^0([0,t_F];H(\nabla\times;\Omega))\cap C^1([0,t_F];\boldsymbol{L}^2(\Omega))$$

(classical semigroup argument, the Maxwell operator being skew-adjoint in the $\boldsymbol{\varepsilon},\boldsymbol{\mu}$-weighted $\boldsymbol{L}^2$ space, see [3]).

2.3. **Structural properties.** The following proposition states that the divergence constraints and the normal magnetic flux are conserved in time.

Proposition 2.1 (Divergence conservation)**.** Let $(\boldsymbol{B},\boldsymbol{E},\boldsymbol{H},\boldsymbol{D})$ be the solution of (2.3)–(2.5), assumed sufficiently regular. Then, for all $t\in[0,t_F]$,

$$\nabla\cdot\boldsymbol{D}(t) = \rho \ \text{ and } \ \nabla\cdot\boldsymbol{B}(t) = 0 \text{ in } \Omega, \qquad \boldsymbol{B}(t)\cdot\boldsymbol{n} = 0 \text{ on } \partial\Omega.$$

Proof. As $\nabla\cdot\nabla\times = 0$, applying the divergence operator to equations (2.3a) and (2.3b) yields

$$\frac{\partial}{\partial t}(\nabla\cdot\boldsymbol{B}) = 0 \ \text{ and } \ \frac{\partial}{\partial t}(\nabla\cdot\boldsymbol{D}) = 0.$$

Moreover, $(\nabla\times\boldsymbol{E})\cdot\boldsymbol{n}$ only depends on the tangential trace $\boldsymbol{n}\times\boldsymbol{E}$, which vanishes by (2.4). Hence the normal component of (2.3a) gives $\frac{\partial}{\partial t}(\boldsymbol{B}\cdot\boldsymbol{n})=0$ on $\partial\Omega$. Integrating in time and using (2.6)–(2.7) concludes the proof. □

A fundamental quantity in the analysis of Maxwell's system (2.3) is the energy $\mathcal{W} : [0,t_F]\to\mathbb{R}$ which is defined for all $t\in[0,t_F]$ by

$$\mathcal{W}(t) := \frac{1}{2}\int_\Omega\big(|\boldsymbol{\varepsilon}^{\frac12}\boldsymbol{E}|^2+|\boldsymbol{\mu}^{\frac12}\boldsymbol{H}|^2\big) = \frac{1}{2}\int_\Omega\big(\boldsymbol{\varepsilon}\boldsymbol{E}\cdot\boldsymbol{E}+\boldsymbol{\mu}\boldsymbol{H}\cdot\boldsymbol{H}\big), \tag{2.9}$$

where $\boldsymbol{\varepsilon}^{\frac12}$ and $\boldsymbol{\mu}^{\frac12}$ are the symmetric positive definite square roots of the tensors. By (2.1),

$$\frac{1}{2}\big(\varepsilon_\flat\|\boldsymbol{E}\|^2_{\boldsymbol{L}^2(\Omega)}+\mu_\flat\|\boldsymbol{H}\|^2_{\boldsymbol{L}^2(\Omega)}\big)\le\mathcal{W}(t)\le\frac{1}{2}\big(\varepsilon_\sharp\|\boldsymbol{E}\|^2_{\boldsymbol{L}^2(\Omega)}+\mu_\sharp\|\boldsymbol{H}\|^2_{\boldsymbol{L}^2(\Omega)}\big),$$

so $\mathcal{W}$ is a quadratic functional of $(\boldsymbol{E},\boldsymbol{H})$ and $(2\mathcal{W})^{1/2}$ is a norm on $\boldsymbol{L}^2(\Omega)\times\boldsymbol{L}^2(\Omega)$, equivalent to the standard one. The energy satisfies the following conservation property.

Proposition 2.2 (Energy conservation)**.** Let $(\boldsymbol{B}, \boldsymbol{E}, \boldsymbol{H}, \boldsymbol{D})$ be the solution of (2.3)–(2.5). Then

$$\mathcal{W}(t) = \mathcal{W}(0), \ \forall t \in [0, t_F].$$

Proof. As $\boldsymbol{\varepsilon}$ and $\boldsymbol{\mu}$ are symmetric and independent of time, and $\boldsymbol{E}, \boldsymbol{H} \in C^1([0, t_F]; \boldsymbol{L}^2(\Omega))$, the map $\mathcal{W}$ is of class C^1 and, by (2.3c)–(2.3d) and (2.3a)–(2.3b),

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{W}(t) = \int_\Omega \frac{\partial}{\partial t}(\boldsymbol{\varepsilon}\boldsymbol{E}) \cdot \boldsymbol{E} + \frac{\partial}{\partial t}(\boldsymbol{\mu}\boldsymbol{H}) \cdot \boldsymbol{H} = \int_\Omega \frac{\partial \boldsymbol{D}}{\partial t} \cdot \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} \cdot \boldsymbol{H} = \int_\Omega \nabla \times \boldsymbol{H} \cdot \boldsymbol{E} - \nabla \times \boldsymbol{E} \cdot \boldsymbol{H}.$$

The Green formula (2.2) with $\boldsymbol{u} = \boldsymbol{E}(t) \in H_0(\nabla\times;\Omega)$ and $\boldsymbol{v} = \boldsymbol{H}(t) \in H(\nabla\times;\Omega)$ gives

$$\int_\Omega \nabla \times \boldsymbol{H} \cdot \boldsymbol{E} - \nabla \times \boldsymbol{E} \cdot \boldsymbol{H} = 0, \tag{2.10}$$

so that $\dfrac{\mathrm{d}}{\mathrm{d}t}\mathcal{W} = 0$ on $[0, t_F]$, and the conclusion follows by integration in time. □

Propositions 2.1 and 2.2 show that, in the study of Maxwell's system (2.3), there are quantities that are conserved in time: the divergences of the fields (and the normal magnetic flux) and the energy of the system. On the one hand, the proof of the conservation of the divergences depends on the underlying *de Rham* structure formed by the differential operators ∇, $\nabla\times$ and $\nabla\cdot$, and particularly on the inclusion

$$\mathrm{Im}(\nabla\times) \subset \mathrm{Ker}(\nabla\cdot). \tag{2.11}$$

On the other hand, the energy conservation relies on *(i)* the symmetry, time independence and uniform positive definiteness (2.1) of the material tensors (so that $\mathcal{W}$ is differentiated as above and controls the fields), and *(ii)* the adjointness relation (2.10).

The CDO framework introduced in the next section is designed to mimic these properties at the discrete level.

3. Discrete setting

3.1. **Meshes.** The domain Ω is partitioned by a primal mesh $\mathrm{M} := \{\mathrm{V}, \mathrm{E}, \mathrm{F}, \mathrm{C}\}$ collecting primal vertices V, primal edges E, primal faces F and primal cells C (possibly polyhedral); since Ω is polyhedral, M fits $\partial\Omega$ exactly. The mesh M is assumed to be an oriented planar cellular complex (see for example [16]) in the sense that *(i)* each closed k-cell is homeomorphic to the closed unit k-ball; *(ii)* the boundary of a cell in C is composed of faces in F, the boundary of a face is composed of edges in E and the boundary of an edge in E is composed of two vertices in V; *(iii)* each entity has an *inner orientation*, which is arbitrarily defined and fixed. In order to isolate the entities that compose a given entity, we define the sets

$$\mathrm{V_e} := \{\mathrm{v} \in \mathrm{V} \mid \mathrm{v} \subset \mathrm{e}\}, \quad \mathrm{E_f} := \{\mathrm{e} \in \mathrm{E} \mid \mathrm{e} \subset \mathrm{f}\}, \quad \mathrm{E_c} := \{\mathrm{e} \in \mathrm{E} \mid \mathrm{e} \subset \mathrm{c}\} \ \text{ and } \ \mathrm{F_c} := \{\mathrm{f} \in \mathrm{F} \mid \mathrm{f} \subset \mathrm{c}\},$$

that collect the two vertices of an edge $\mathrm{e} \in \mathrm{E}$, the edges of a face $\mathrm{f} \in \mathrm{F}$, and the edges and the faces of a cell $\mathrm{c} \in \mathrm{C}$, respectively. Conversely, the sets

$$\mathrm{F_e} := \{\mathrm{f} \in \mathrm{F} \mid \mathrm{e} \subset \mathrm{f}\} \ \text{ and } \ \mathrm{C_f} := \{\mathrm{c} \in \mathrm{C} \mid \mathrm{f} \subset \mathrm{c}\},$$

collect the faces in F that contain the edge e and the cells in C that contain the face f.

In addition to the primal mesh, we introduce a dual mesh $\widetilde{\mathrm{M}} := \{\widetilde{\mathrm{V}}, \widetilde{\mathrm{E}}, \widetilde{\mathrm{F}}, \widetilde{\mathrm{C}}\}$ defined by a one-to-one pairing with the entities of the primal mesh M as follows: to each primal vertex $\mathrm{v} \in \mathrm{V}$ we associate a dual cell $\tilde{\mathrm{c}}(\mathrm{v}) \in \widetilde{\mathrm{C}}$, to each primal edge $\mathrm{e} \in \mathrm{E}$ we associate a dual face $\tilde{\mathrm{f}}(\mathrm{e}) \in \widetilde{\mathrm{F}}$, to each primal face we associate a dual edge $\tilde{\mathrm{e}}(\mathrm{f}) \in \widetilde{\mathrm{E}}$ and to each primal cell $\mathrm{c} \in \mathrm{C}$ we associate a dual vertex $\tilde{\mathrm{v}}(\mathrm{c}) \in \widetilde{\mathrm{V}}$. With this definition, the dual mesh is not necessarily a planar cellular complex as some entities can be broken (piecewise straight edges, piecewise planar faces). However, each entity in the dual mesh has an orientation that is induced by its corresponding entity in the primal mesh. For clarity, denoting by $\boldsymbol{n}_\mathrm{x}$ the unit normal of a face x (either primal or dual) and by $\boldsymbol{\tau}_\mathrm{y}$ the unit directional vector of an edge y (either primal or dual) we assume that the relations

$$\boldsymbol{n}_\mathrm{f} \cdot \boldsymbol{\tau}_{\tilde{\mathrm{e}}(\mathrm{f})} > 0 \ \text{ and } \ \boldsymbol{\tau}_\mathrm{e} \cdot \boldsymbol{n}_{\tilde{\mathrm{f}}(\mathrm{e})} > 0 \tag{3.1}$$

hold for all $\mathrm{f} \in \mathrm{F}$ and $\mathrm{e} \in \mathrm{E}$. For a broken dual edge or face, the tangent or the normal is defined on each straight segment or planar piece, and (3.1) is assumed on each piece.

Practically speaking, the dual mesh can be built in different ways, for example using Voronoi diagrams. In the following, the analysis of the CDO scheme is derived using a *barycentric* dual mesh. For the latter to be well defined, we assume that the entities of M are star-shaped with respect to their barycenter (the barycenter of a vertex being itself). For $\mathrm{c} \in \mathrm{C}$ we write $h_\mathrm{c} := \mathrm{diam}(\mathrm{c})$, $h := \max_{\mathrm{c}\in\mathrm{C}} h_\mathrm{c}$ and $h_{\min} := \min_{\mathrm{c}\in\mathrm{C}} h_\mathrm{c}$. The analysis considers a sequence of such meshes with $h \to 0$. In addition, we assume that there exists a simplicial submesh

common to M and $\widetilde{\mathrm{M}}$, namely the barycentric subdivision of M (assumed nondegenerate), which is shape-regular uniformly over the mesh family and contains a uniformly bounded number of simplices in each primal cell.

To discretize the time interval $[0, t_F]$, let $N \geq 2$ be an integer, $\Delta t := t_F/N$ the time step and $t_n := n\Delta t$ for $n \in [\![0, N]\!]$ the *time instants*. The *primal time mesh* $\mathrm{T} := \{\mathrm{I}, \mathrm{P}\}$ is composed of the set of time instants $\mathrm{I} := \{t_n\}_{0\leq n\leq N}$ and the set of *time intervals* $\mathrm{P} := \{[t_n, t_{n+1}]\}_{0\leq n\leq N-1}$. Assigning to its entities the canonical orientation of $\mathbb{R}$, T is a 1D oriented cellular complex: each interval in P is bounded by two time instants in I.

As for the spatial mesh, the *dual time mesh* $\widetilde{\mathrm{T}} := \{\widetilde{\mathrm{I}}, \widetilde{\mathrm{P}}\}$ is defined by a one-to-one pairing. To each primal time interval $[t_n, t_{n+1}]$ we associate its midpoint $t_{n+\frac{1}{2}} := (t_n + t_{n+1})/2$, and to each primal time instant t_n we associate the dual time interval

$$J_n := [t_{n-\frac{1}{2}}, t_{n+\frac{1}{2}}], \quad n \in [\![0, N]\!],$$

with the convention $t_{-1/2} := t_0$ and $t_{N+1/2} := t_N$: the dual intervals attached to the initial and final instants are half intervals, closed by t_0 and t_N. Their lengths are $|J_n| = \Delta t$ for $1 \leq n \leq N-1$ and $|J_0| = |J_N| = \Delta t/2$.

3.2. **Degrees of freedom.** In the CDO scheme for Maxwell's equations (2.3) the discrete unknowns are attached to edges and faces. Precisely, those corresponding to $\boldsymbol{E}$ are attached to primal edges and those corresponding to $\boldsymbol{H}$ to dual edges, while the unknowns associated to $\boldsymbol{B}$ and $\boldsymbol{D}$ are attached to primal and dual faces respectively. The spaces collecting the degrees of freedom (dofs) on the primal mesh are denoted by $\mathcal{V}$, $\mathcal{E}$, $\mathcal{F}$ and $\mathcal{C}$. The latter are defined as the codomains of the *reduction* maps acting as follows:

$$\mathsf{R}_{\mathcal{V}}(p)|_{\mathrm{v}} := p(\boldsymbol{x}_{\mathrm{v}}), \qquad \forall \mathrm{v} \in \mathrm{V}, \qquad \mathsf{R}_{\mathcal{E}}(\boldsymbol{\psi})|_{\mathrm{e}} := \int_{\mathrm{e}} \boldsymbol{\psi} \cdot \boldsymbol{\tau}_{\mathrm{e}}, \quad \forall \mathrm{e} \in \mathrm{E}, \tag{3.2a}$$

$$\mathsf{R}_{\mathcal{F}}(\boldsymbol{\varphi})|_{\mathrm{f}} := \int_{\mathrm{f}} \boldsymbol{\varphi} \cdot \boldsymbol{n}_{\mathrm{f}}, \quad \forall \mathrm{f} \in \mathrm{F}, \qquad \mathsf{R}_{\mathcal{C}}(w)|_{\mathrm{c}} := \int_{\mathrm{c}} w, \qquad \forall \mathrm{c} \in \mathrm{C}, \tag{3.2b}$$

where p, $\boldsymbol{\psi}$, $\boldsymbol{\varphi}$ and w are fields regular enough for these point values and integrals to be well-defined (for instance continuous fields); the identities involving reduction maps below are understood for such fields. Thus each space contains one real number per mesh entity: $\mathcal{V} = \mathbb{R}^{\mathrm{V}}$, $\mathcal{E} = \mathbb{R}^{\mathrm{E}}$, $\mathcal{F} = \mathbb{R}^{\mathrm{F}}$ and $\mathcal{C} = \mathbb{R}^{\mathrm{C}}$. The spaces $\widetilde{\mathcal{V}}$, $\widetilde{\mathcal{E}}$, $\widetilde{\mathcal{F}}$ and $\widetilde{\mathcal{C}}$ of dofs on the dual mesh are defined in the same way, using reduction maps analogous to (3.2); integrals over broken dual entities are sums over their pieces. By the one-to-one pairing between primal and dual entities, a dual dof is attached to a primal entity of complementary dimension: $\widetilde{\mathcal{C}}$ has one entry per primal vertex, $\widetilde{\mathcal{F}}$ one per primal edge, $\widetilde{\mathcal{E}}$ one per primal face and $\widetilde{\mathcal{V}}$ one per primal cell. This pairing is what the duality products below rely on. Following the usual terminology, the dofs attached to edges are called *circulations* and those attached to faces *fluxes*.

The time dofs are defined in the same spirit. For a continuous function u on $[0, t_F]$, the reduction maps on the primal and dual time instants are

$$u \mapsto \big(u(t_n)\big)_{0\leq n\leq N} \quad \text{and} \quad u \mapsto \big(u(t_{n+\frac{1}{2}})\big)_{-1\leq n\leq N},$$

and the reduction maps on the primal and dual time intervals are

$$u \mapsto \Big(\int_{t_n}^{t_{n+1}} u\Big)_{0\leq n\leq N-1} \quad \text{and} \quad u \mapsto \Big(\int_{J_n} u\Big)_{0\leq n\leq N}.$$

3.3. **Discrete differential operators.** The discrete differential operators on the primal mesh $\mathsf{GRAD} : \mathcal{V} \to \mathcal{E}$, $\mathsf{ROT} : \mathcal{E} \to \mathcal{F}$ and $\mathsf{DIV} : \mathcal{F} \to \mathcal{C}$ are defined such that

$$\forall \mathsf{p} \in \mathcal{V}, \qquad \mathsf{GRAD}(\mathsf{p})_{\mathrm{e}} := \sum_{\mathrm{v}\in\mathrm{V}_{\mathrm{e}}} \iota_{\mathrm{v,e}} \mathsf{p}_{\mathrm{v}} \qquad \forall \mathrm{e} \in \mathrm{E}, \tag{3.3a}$$

$$\forall \boldsymbol{\psi} \in \mathcal{E}, \qquad \mathsf{ROT}(\boldsymbol{\psi})_{\mathrm{f}} := \sum_{\mathrm{e}\in\mathrm{E}_{\mathrm{f}}} \iota_{\mathrm{e,f}} \boldsymbol{\psi}_{\mathrm{e}} \qquad \forall \mathrm{f} \in \mathrm{F}, \tag{3.3b}$$

$$\forall \boldsymbol{\phi} \in \mathcal{F}, \qquad \mathsf{DIV}(\boldsymbol{\phi})_{\mathrm{c}} := \sum_{\mathrm{f}\in\mathrm{F}_{\mathrm{c}}} \iota_{\mathrm{f,c}} \boldsymbol{\phi}_{\mathrm{f}} \qquad \forall \mathrm{c} \in \mathrm{C}, \tag{3.3c}$$

with $\iota_{\mathrm{v,e}} = +1$ if $\boldsymbol{\tau}_{\mathrm{e}}$ points toward v and $\iota_{\mathrm{v,e}} = -1$ otherwise; $\iota_{\mathrm{e,f}} = +1$ if $\boldsymbol{\tau}_{\mathrm{e}}$ shares the same orientation as the one induced by $\boldsymbol{n}_{\mathrm{f}}$ and $\iota_{\mathrm{e,f}} = -1$ otherwise; $\iota_{\mathrm{f,c}} = +1$ if $\boldsymbol{n}_{\mathrm{f}}$ points outward from c and $\iota_{\mathrm{f,c}} = -1$ otherwise. These operators depend only on the orientation of the entities of the primal mesh and are algebraically represented by sparse matrices with entries in $\{0, \pm 1\}$. These definitions lead to two important properties: the first is the discrete counterpart of (2.11), and the second states that these operators commute with the reduction maps (3.2).

Proposition 3.1 (Complex property)**.**

$$\mathrm{Im}(\mathsf{GRAD}) \subset \mathrm{Ker}(\mathsf{ROT}), \quad \mathrm{Im}(\mathsf{ROT}) \subset \mathrm{Ker}(\mathsf{DIV}).$$

Proposition 3.2 (Commuting property)**.**

$$\mathsf{R}_{\mathcal{E}}(\nabla) = \mathsf{GRAD}(\mathsf{R}_{\mathcal{V}}), \quad \mathsf{R}_{\mathcal{F}}(\nabla\times) = \mathsf{ROT}(\mathsf{R}_{\mathcal{E}}) \ \text{ and } \ \mathsf{R}_{\mathcal{C}}(\nabla\cdot) = \mathsf{DIV}(\mathsf{R}_{\mathcal{F}}).$$

Both properties, as well as their dual counterparts below, follow from the Stokes formula applied entity by entity, see for example [7].

As mentioned above, the orientation on the dual mesh is induced by the one fixed on the primal mesh. This is expressed by the incidence relations

$$\iota_{\tilde{\mathrm{v}}(\mathrm{c}),\tilde{\mathrm{e}}(\mathrm{f})} := -\iota_{\mathrm{f,c}}, \quad \iota_{\tilde{\mathrm{e}}(\mathrm{f}),\tilde{\mathrm{f}}(\mathrm{e})} := \iota_{\mathrm{e,f}} \ \text{ and } \ \iota_{\tilde{\mathrm{f}}(\mathrm{e}),\tilde{\mathrm{c}}(\mathrm{v})} := -\iota_{\mathrm{v,e}}. \tag{3.4}$$

Hence, the discrete differential operators on the dual mesh $\widetilde{\mathsf{GRAD}} : \widetilde{\mathcal{V}} \to \widetilde{\mathcal{E}}$, $\widetilde{\mathsf{ROT}} : \widetilde{\mathcal{E}} \to \widetilde{\mathcal{F}}$ and $\widetilde{\mathsf{DIV}} : \widetilde{\mathcal{F}} \to \widetilde{\mathcal{C}}$ can be defined in the same way as in (3.3). Defining the duality products

$$[\![\mathsf{u},\boldsymbol{\rho}]\!]_{\mathcal{V}\widetilde{\mathcal{C}}} := \sum_{\mathrm{v}\in\mathrm{V}} \mathsf{u}_{\mathrm{v}}\boldsymbol{\rho}_{\tilde{\mathrm{c}}(\mathrm{v})}, \ \forall \mathsf{u}\in\mathcal{V}, \ \forall\boldsymbol{\rho}\in\widetilde{\mathcal{C}}, \qquad [\![\mathsf{g},\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} := \sum_{\mathrm{e}\in\mathrm{E}} \mathsf{g}_{\mathrm{e}}\boldsymbol{\phi}_{\tilde{\mathrm{f}}(\mathrm{e})}, \ \forall \mathsf{g}\in\mathcal{E}, \ \forall\boldsymbol{\phi}\in\widetilde{\mathcal{F}},$$
$$[\![\boldsymbol{\phi},\mathsf{g}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} := \sum_{\mathrm{f}\in\mathrm{F}} \boldsymbol{\phi}_{\mathrm{f}}\mathsf{g}_{\tilde{\mathrm{e}}(\mathrm{f})}, \ \forall\boldsymbol{\phi}\in\mathcal{F}, \ \forall \mathsf{g}\in\widetilde{\mathcal{E}}, \qquad [\![\boldsymbol{\rho},\mathsf{u}]\!]_{\mathcal{C}\widetilde{\mathcal{V}}} := \sum_{\mathrm{c}\in\mathrm{C}} \boldsymbol{\rho}_{\mathrm{c}}\mathsf{u}_{\tilde{\mathrm{v}}(\mathrm{c})}, \ \forall\boldsymbol{\rho}\in\mathcal{C}, \ \forall \mathsf{u}\in\widetilde{\mathcal{V}},$$

they are linked to the operators on the primal mesh by the following adjunction relations:

$$[\![\mathsf{p},\widetilde{\mathsf{DIV}}(\mathsf{u})]\!]_{\mathcal{V}\widetilde{\mathcal{C}}} = -[\![\mathsf{GRAD}(\mathsf{p}),\mathsf{u}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}, \quad \forall(\mathsf{p},\mathsf{u})\in\mathcal{V}\times\widetilde{\mathcal{F}}, \tag{3.5a}$$
$$[\![\mathsf{g},\widetilde{\mathsf{ROT}}(\mathsf{v})]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} = +[\![\mathsf{ROT}(\mathsf{g}),\mathsf{v}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}, \quad \forall(\mathsf{g},\mathsf{v})\in\mathcal{E}\times\widetilde{\mathcal{E}}, \tag{3.5b}$$
$$[\![\mathsf{u},\widetilde{\mathsf{GRAD}}(\mathsf{p})]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} = -[\![\mathsf{DIV}(\mathsf{u}),\mathsf{p}]\!]_{\mathcal{C}\widetilde{\mathcal{V}}}, \quad \forall(\mathsf{u},\mathsf{p})\in\mathcal{F}\times\widetilde{\mathcal{V}}. \tag{3.5c}$$

Note that (3.5b) is the discrete counterpart of (2.10). From these relations, the matrices of the dual operators are the transposes (up to a sign) of those of the primal operators. As on the primal mesh, we have two important properties: Proposition 3.3 is the dual version of Proposition 3.1, and Proposition 3.4 states that the dual reduction maps commute with the dual operators only on dual entities that do not touch the boundary.

Proposition 3.3 (Dual complex property)**.**

$$\mathrm{Im}(\widetilde{\mathsf{GRAD}}) \subset \mathrm{Ker}(\widetilde{\mathsf{ROT}}), \quad \mathrm{Im}(\widetilde{\mathsf{ROT}}) \subset \mathrm{Ker}(\widetilde{\mathsf{DIV}}).$$

Proposition 3.4 (Partial commuting property)**.**

$$\mathsf{R}_{\widetilde{\mathcal{E}}}(\nabla)|_{\tilde{\mathrm{e}}(\mathrm{f})} = \widetilde{\mathsf{GRAD}}(\mathsf{R}_{\widetilde{\mathcal{V}}})|_{\tilde{\mathrm{e}}(\mathrm{f})}, \quad \forall \mathrm{f}\in\mathrm{F}\setminus\mathrm{F}^{\partial},$$
$$\mathsf{R}_{\widetilde{\mathcal{F}}}(\nabla\times)|_{\tilde{\mathrm{f}}(\mathrm{e})} = \widetilde{\mathsf{ROT}}(\mathsf{R}_{\widetilde{\mathcal{E}}})|_{\tilde{\mathrm{f}}(\mathrm{e})}, \quad \forall \mathrm{e}\in\mathrm{E}\setminus\mathrm{E}^{\partial},$$
$$\mathsf{R}_{\widetilde{\mathcal{C}}}(\nabla\cdot)|_{\tilde{\mathrm{c}}(\mathrm{v})} = \widetilde{\mathsf{DIV}}(\mathsf{R}_{\widetilde{\mathcal{F}}})|_{\tilde{\mathrm{c}}(\mathrm{v})}, \quad \forall \mathrm{v}\in\mathrm{V}\setminus\mathrm{V}^{\partial},$$

with V^{∂}, E^{∂} and F^{∂} the sets collecting the vertices, the edges and the faces that lie on the boundary, respectively.

The restriction to interior entities comes from the truncation of the dual mesh by $\partial\Omega$: for a dual entity attached to a boundary primal entity, the part of its boundary lying on $\partial\Omega$ is not covered by dual entities, so its contribution in the Stokes formula is missing from the dual operator.

3.4. **Discrete Hodge operators.** The cornerstone of CDO schemes is the definition of material-dependent discrete operators that encode the metric. We call them *discrete Hodge operators*, by analogy with the Hodge star operator: they map dofs attached to the primal mesh to dofs attached to the dual mesh. As in finite element methods, they are assembled cell by cell.

For a cell $\mathrm{c}\in\mathrm{C}$, we denote by $\mathcal{E}_{\mathrm{c}}$, $\mathcal{F}_{\mathrm{c}}$, $\widetilde{\mathcal{E}}_{\mathrm{c}}$ and $\widetilde{\mathcal{F}}_{\mathrm{c}}$ the spaces collecting the local dofs and by $\mathsf{R}_{\mathcal{E}_c}$, $\mathsf{R}_{\mathcal{F}_c}$, $\mathsf{R}_{\widetilde{\mathcal{E}}_c}$ and $\mathsf{R}_{\widetilde{\mathcal{F}}_c}$ the corresponding local reduction maps. The local duality products are defined by

$$[\![\mathsf{g},\boldsymbol{\psi}]\!]_{\mathcal{E}_c\widetilde{\mathcal{F}}_c} := \sum_{\mathrm{e}\in\mathrm{E}_c} \mathsf{g}_{\mathrm{e}}\boldsymbol{\psi}_{\tilde{\mathrm{f}}_{\mathrm{c}}(\mathrm{e})}, \ \forall \mathsf{g}\in\mathcal{E}_{\mathrm{c}}, \ \forall\boldsymbol{\psi}\in\widetilde{\mathcal{F}}_{\mathrm{c}}, \qquad [\![\boldsymbol{\phi},\mathsf{h}]\!]_{\mathcal{F}_c\widetilde{\mathcal{E}}_c} := \sum_{\mathrm{f}\in\mathrm{F}_c} \boldsymbol{\phi}_{\mathrm{f}}\mathsf{h}_{\tilde{\mathrm{e}}_{\mathrm{c}}(\mathrm{f})}, \ \forall\boldsymbol{\phi}\in\mathcal{F}_{\mathrm{c}}, \ \forall \mathsf{h}\in\widetilde{\mathcal{E}}_{\mathrm{c}},$$

where $\tilde{\mathrm{f}}_{\mathrm{c}}(\mathrm{e}) := \tilde{\mathrm{f}}(\mathrm{e})\cap\mathrm{c}$ and $\tilde{\mathrm{e}}_{\mathrm{c}}(\mathrm{f}) := \tilde{\mathrm{e}}(\mathrm{f})\cap\mathrm{c}$. Using these notations, let

$$\mathrm{H}_{\varepsilon}^{\mathcal{E}_c\widetilde{\mathcal{F}}_c} : \mathcal{E}_{\mathrm{c}} \to \widetilde{\mathcal{F}}_{\mathrm{c}} \ \text{ and } \ \mathrm{H}_{\mu^{-1}}^{\mathcal{F}_c\widetilde{\mathcal{E}}_c} : \mathcal{F}_{\mathrm{c}} \to \widetilde{\mathcal{E}}_{\mathrm{c}},$$

be two discrete operators that verify the following properties.

(**H0c**) $\mathrm{H}_{\varepsilon}^{\mathcal{E}_c\widetilde{\mathcal{F}}_c}$ and $\mathrm{H}_{\mu^{-1}}^{\mathcal{F}_c\widetilde{\mathcal{E}}_c}$ are symmetric,

(**H1c**) Denoting by $\varepsilon_{\flat,\mathrm{c}}$ and $\varepsilon_{\sharp,\mathrm{c}}$ (resp. $\mu_{\flat,\mathrm{c}}$ and $\mu_{\sharp,\mathrm{c}}$) the essential lower and upper eigenvalue bounds of $\boldsymbol{\varepsilon}$ (resp. $\boldsymbol{\mu}$) in c, there exist constants $\eta_{\varepsilon,\mathrm{c}}, \eta_{\mu^{-1},\mathrm{c}} > 0$, bounded uniformly in c and over the mesh family, such that

$$\eta_{\varepsilon,\mathrm{c}}^{-1}\varepsilon_{\flat,\mathrm{c}}\|\mathsf{g}\|_{\mathcal{E},\mathrm{c}}^2 \le [\![\mathsf{g}, \mathrm{H}_{\varepsilon}^{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}\mathsf{g}]\!]_{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}} \le \eta_{\varepsilon,\mathrm{c}}\varepsilon_{\sharp,\mathrm{c}}\|\mathsf{g}\|_{\mathcal{E},\mathrm{c}}^2, \qquad \forall \mathsf{g} \in \mathcal{E}_\mathrm{c}, \tag{3.6a}$$

$$\eta_{\mu^{-1},\mathrm{c}}^{-1}\mu_{\sharp,\mathrm{c}}^{-1}\|\boldsymbol{\phi}\|_{\mathcal{F},\mathrm{c}}^2 \le [\![\boldsymbol{\phi}, \mathrm{H}_{\mu^{-1}}^{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}}\boldsymbol{\phi}]\!]_{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}} \le \eta_{\mu^{-1},\mathrm{c}}\mu_{\flat,\mathrm{c}}^{-1}\|\boldsymbol{\phi}\|_{\mathcal{F},\mathrm{c}}^2, \ \forall \boldsymbol{\phi} \in \mathcal{F}_\mathrm{c}, \tag{3.6b}$$

with $\|\bullet\|_{\mathcal{E},\mathrm{c}}^2 := \sum_{\mathrm{e}\in\mathrm{E}_\mathrm{c}} h_\mathrm{c}(\bullet)^2$ and $\|\bullet\|_{\mathcal{F},\mathrm{c}}^2 := \sum_{\mathrm{f}\in\mathrm{F}_\mathrm{c}} h_\mathrm{c}^{-1}(\bullet)^2$.

(**H2c**) For every constant field $\boldsymbol{K} \in [\mathbb{P}_0(\mathrm{c})]^3$ the following relations hold:

$$\mathrm{H}_{\varepsilon}^{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}\mathsf{R}_{\mathcal{E}_c}(\boldsymbol{K}) - \mathsf{R}_{\widetilde{\mathcal{F}}_c}(\boldsymbol{\varepsilon}\boldsymbol{K}) = 0, \tag{3.7}$$

$$\mathrm{H}_{\mu^{-1}}^{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}}\mathsf{R}_{\mathcal{F}_c}(\boldsymbol{K}) - \mathsf{R}_{\widetilde{\mathcal{E}}_c}(\boldsymbol{\mu}^{-1}\boldsymbol{K}) = 0. \tag{3.8}$$

Property (**H2c**) states that the local Hodge operators are exact for constant fields. Local operators satisfying (**H0c**)–(**H2c**) on polyhedral meshes are constructed for instance in [6, 20].

Denoting by $\mathsf{P}_{\mathcal{E}_\mathrm{c}} : \mathcal{E} \to \mathcal{E}_\mathrm{c}$ and $\mathsf{P}_{\mathcal{F}_\mathrm{c}} : \mathcal{F} \to \mathcal{F}_\mathrm{c}$ the restriction maps from global dofs to local dofs, the global Hodge operators

$$\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}} : \mathcal{E} \to \widetilde{\mathcal{F}} \ \text{ and } \ \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} : \mathcal{F} \to \widetilde{\mathcal{E}},$$

are defined by

$$[\![\mathsf{g}, \mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}}\boldsymbol{\psi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} := \sum_{\mathrm{c}\in\mathrm{C}} [\![\mathsf{P}_{\mathcal{E}_\mathrm{c}}(\mathsf{g}), \mathrm{H}_{\varepsilon}^{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}\mathsf{P}_{\mathcal{E}_\mathrm{c}}(\boldsymbol{\psi})]\!]_{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}, \quad \forall \mathsf{g}, \boldsymbol{\psi} \in \mathcal{E}, \tag{3.9a}$$

$$[\![\boldsymbol{\phi}, \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} := \sum_{\mathrm{c}\in\mathrm{C}} [\![\mathsf{P}_{\mathcal{F}_\mathrm{c}}(\boldsymbol{\phi}), \mathrm{H}_{\mu^{-1}}^{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}}\mathsf{P}_{\mathcal{F}_\mathrm{c}}(\boldsymbol{\psi})]\!]_{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}}, \ \forall \boldsymbol{\phi}, \boldsymbol{\psi} \in \mathcal{F}. \tag{3.9b}$$

In what follows, we omit the restriction maps and simply write $[\![\mathsf{g}, \mathrm{H}_{\varepsilon}^{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}\boldsymbol{\psi}]\!]_{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}$, the local character being carried by the subscript c of the duality product.

Summing the local bounds of (**H1c**) over the cells, and using $\varepsilon_\flat \le \varepsilon_{\flat,\mathrm{c}}$, $\varepsilon_{\sharp,\mathrm{c}} \le \varepsilon_\sharp$ (and similarly for $\boldsymbol{\mu}$) with the bounds of (2.1), the global Hodge operators satisfy:

(**H1**) With $\eta_\varepsilon := \max_{\mathrm{c}\in\mathrm{C}} \eta_{\varepsilon,\mathrm{c}}$ and $\eta_{\mu^{-1}} := \max_{\mathrm{c}\in\mathrm{C}} \eta_{\mu^{-1},\mathrm{c}}$, which are bounded independently of h,

$$\eta_\varepsilon^{-1}\varepsilon_\flat\|\mathsf{g}\|_{2,\mathcal{E}}^2 \le [\![\mathsf{g}, \mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{g}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} \le \eta_\varepsilon\varepsilon_\sharp\|\mathsf{g}\|_{2,\mathcal{E}}^2, \qquad \forall \mathsf{g} \in \mathcal{E}, \tag{3.10a}$$

$$\eta_{\mu^{-1}}^{-1}\mu_\sharp^{-1}\|\boldsymbol{\phi}\|_{2,\mathcal{F}}^2 \le [\![\boldsymbol{\phi}, \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\boldsymbol{\phi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} \le \eta_{\mu^{-1}}\mu_\flat^{-1}\|\boldsymbol{\phi}\|_{2,\mathcal{F}}^2, \ \forall \boldsymbol{\phi} \in \mathcal{F}, \tag{3.10b}$$

with $\|\bullet\|_{2,\mathcal{E}}^2 := \sum_{\mathrm{c}\in\mathrm{C}}\|\bullet\|_{\mathcal{E},\mathrm{c}}^2$ and $\|\bullet\|_{2,\mathcal{F}}^2 := \sum_{\mathrm{c}\in\mathrm{C}}\|\bullet\|_{\mathcal{F},\mathrm{c}}^2$.

We now have all the ingredients to state the CDO scheme for Maxwell's equations (2.3).

4. CDO scheme

The CDO scheme is a leapfrog time integration of the discrete Faraday and Ampère laws, the constitutive relations being enforced through the Hodge operators. We first describe how the PEC condition is taken into account. The PEC condition (2.4) is imposed strongly on the electric unknowns, which lie in

$$\mathcal{E}_0 := \{\mathsf{g} \in \mathcal{E} : \mathsf{g}_\mathrm{e} = 0 \ \forall \mathrm{e} \in \mathrm{E}^\partial\}.$$

All the other unknowns live on the full primal and dual meshes. Since the boundary values of the electric unknowns are prescribed, the constitutive relation (2.3c) is only imposed on the dual faces attached to interior edges: for $\boldsymbol{\phi} \in \widetilde{\mathcal{F}}$, we denote by $(\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}$ the unique $\mathsf{g} \in \mathcal{E}_0$ such that

$$[\![\mathsf{g}', \mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{g}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} = [\![\mathsf{g}', \boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}, \quad \forall \mathsf{g}' \in \mathcal{E}_0. \tag{4.1}$$

This map is well defined since, by (**H0c**) and (**H1**), $\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}}$ restricted to $\mathcal{E}_0$ is symmetric positive definite. In matrix form, one solves with the submatrix of $\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}}$ associated with interior edges and sets the boundary entries to zero.

The CDO scheme for problem (2.3) reads: given $\mathsf{D}^0 = \mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}_0)$ and $\mathsf{B}^0 = \mathsf{R}_{\mathcal{F}}(\boldsymbol{B}_0)$, set

$$\mathsf{E}^0 = \frac{\Delta t}{2}(\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{D}^0, \tag{4.2}$$

then compute

$$\mathsf{B}^{n+\frac{1}{2}} - \mathsf{B}^{n-\frac{1}{2}} + \mathsf{ROT}(\mathsf{E}^n) = 0, \quad \forall n \in [\![0, N-1]\!], \tag{4.3a}$$

$$\mathsf{H}^{n+\frac{1}{2}} = \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{B}^{n+\frac{1}{2}}, \quad \forall n \in [\![0, N-1]\!], \tag{4.3b}$$

$$\mathsf{D}^{n+1} - \mathsf{D}^n - \widetilde{\mathsf{ROT}}(\mathsf{H}^{n+\frac{1}{2}}) = 0, \quad \forall n \in [\![0, N-1]\!], \tag{4.3c}$$

$$\mathsf{E}^{n+1} = \Delta t (\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1} \mathsf{D}^{n+1}, \quad \forall n \in [\![0, N-2]\!], \tag{4.3d}$$

with the convention $\mathsf{B}^{-\frac{1}{2}} := \mathsf{B}^0$ in (4.3a) for $n = 0$. Finally, the unknowns E^N, B^N and H^N are given by

$$\mathsf{E}^N = \frac{\Delta t}{2} (\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1} \mathsf{D}^N, \quad \mathsf{B}^N = \mathsf{B}^{N-\frac{1}{2}} - \mathsf{ROT}(\mathsf{E}^N), \qquad \mathsf{H}^N = \frac{\Delta t}{2} \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{B}^N. \tag{4.4}$$

By construction of $(\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}$, the electric unknowns satisfy the discrete boundary condition

$$\mathsf{E}_{\mathrm{e}}^n = 0, \ \forall \mathrm{e} \in \mathrm{E}^{\partial}, \ \forall n \in [\![0, N]\!]. \tag{4.5}$$

Although H^0 is not used in the iterations (4.3), it is convenient for the analysis to define it by

$$\mathsf{H}^0 = \frac{\Delta t}{2} \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{B}^0. \tag{4.6}$$

The factors Δt and $\Delta t/2$ in (4.2)–(4.6) are the lengths of the time intervals on which E and H live. Indeed, following the time dofs of Section 3.2, D^n and $\mathsf{B}^{n+\frac{1}{2}}$ approximate the instantaneous fluxes $\mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_n))$ and $\mathsf{R}_{\mathcal{F}}(\boldsymbol{B}(t_{n+\frac{1}{2}}))$, whereas E^n and $\mathsf{H}^{n+\frac{1}{2}}$ approximate the circulations integrated in time $\int_{J_n} \mathsf{R}_{\mathcal{E}}(\boldsymbol{E})$ and $\int_{t_n}^{t_{n+1}} \mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H})$ (over the half intervals $[t_0, t_{\frac{1}{2}}]$ and $[t_{N-\frac{1}{2}}, t_N]$ for H^0 and H^N). Hence the time averages of the fields are $\mathsf{E}^n/|J_n|$ and $\mathsf{H}^{n+\frac{1}{2}}/\Delta t$. Each time step only involves sparse matrix-vector products, except for the electric update, which requires a solve with the symmetric positive definite matrix of (4.1).

The following proposition is the discrete counterpart of Proposition 2.1: the discrete divergences and the boundary magnetic fluxes are conserved by the scheme (4.2)–(4.4).

Proposition 4.1 (Discrete divergence conservation)**.** Let $(\mathsf{B}^{n+\frac{1}{2}})_n$, B^N and $(\mathsf{D}^n)_n$ be given by (4.2)–(4.4). The magnetic fluxes $\mathsf{B}^{n+\frac{1}{2}}$, $n \in [\![0, N-1]\!]$, and B^N are divergence-free and vanish on the boundary faces:

$$\mathsf{DIV}(\mathsf{B}^{n+\frac{1}{2}}) = \mathsf{DIV}(\mathsf{B}^N) = 0 \ \text{ and } \ \mathsf{B}_{\mathrm{f}}^{n+\frac{1}{2}} = \mathsf{B}_{\mathrm{f}}^N = 0, \ \forall \mathrm{f} \in \mathrm{F}^{\partial}.$$

Moreover, for all $n \in [\![0, N]\!]$ and all $\mathrm{v} \in \mathrm{V} \setminus \mathrm{V}^{\partial}$,

$$\widetilde{\mathsf{DIV}}(\mathsf{D}^n)|_{\tilde{\mathrm{c}}(\mathrm{v})} = \mathsf{R}_{\widetilde{\mathcal{C}}}(\rho)|_{\tilde{\mathrm{c}}(\mathrm{v})}.$$

Proof. By (2.6) and the commuting properties of Propositions 3.2 and 3.4,

$$\mathsf{DIV}(\mathsf{B}^0) = \mathsf{R}_{\mathcal{C}}(\nabla \cdot \boldsymbol{B}_0) = 0 \ \text{ and } \ \widetilde{\mathsf{DIV}}(\mathsf{D}^0)|_{\tilde{\mathrm{c}}(\mathrm{v})} = \mathsf{R}_{\widetilde{\mathcal{C}}}(\rho)|_{\tilde{\mathrm{c}}(\mathrm{v})}, \ \forall \mathrm{v} \in \mathrm{V} \setminus \mathrm{V}^{\partial}.$$

Applying DIV to (4.3a) and to the second relation of (4.4), and using $\mathsf{DIV}\,\mathsf{ROT} = 0$ from Proposition 3.1, gives $\mathsf{DIV}(\mathsf{B}^{n+\frac{1}{2}}) = \mathsf{DIV}(\mathsf{B}^{n-\frac{1}{2}})$ for $n \in [\![0, N-1]\!]$ and $\mathsf{DIV}(\mathsf{B}^N) = \mathsf{DIV}(\mathsf{B}^{N-\frac{1}{2}})$. By induction, all these divergences equal $\mathsf{DIV}(\mathsf{B}^{-\frac{1}{2}}) = \mathsf{DIV}(\mathsf{B}^0) = 0$. Similarly, for $\mathrm{f} \in \mathrm{F}^{\partial}$, all the edges of f belong to E^{∂}, so that $\mathsf{ROT}(\mathsf{E}^n)_{\mathrm{f}} = 0$ by (4.5). Hence (4.3a) and (4.4) keep B_{f} equal to $\mathsf{B}_{\mathrm{f}}^0 = \int_{\mathrm{f}} \boldsymbol{B}_0 \cdot \boldsymbol{n}_{\mathrm{f}} = 0$, by (2.7).

In the same way, applying $\widetilde{\mathsf{DIV}}$ to (4.3c) and using $\widetilde{\mathsf{DIV}}\,\widetilde{\mathsf{ROT}} = 0$ from Proposition 3.3 gives $\widetilde{\mathsf{DIV}}(\mathsf{D}^{n+1}) = \widetilde{\mathsf{DIV}}(\mathsf{D}^n)$ for $n \in [\![0, N-1]\!]$, whence $\widetilde{\mathsf{DIV}}(\mathsf{D}^n) = \widetilde{\mathsf{DIV}}(\mathsf{D}^0)$ for all $n \in [\![0, N]\!]$, and the result follows from the initial identity on the dual cells $\tilde{\mathrm{c}}(\mathrm{v})$, $\mathrm{v} \in \mathrm{V} \setminus \mathrm{V}^{\partial}$. □

The electric divergence is only controlled on the dual cells attached to interior vertices: the dual cells attached to boundary vertices are truncated by $\partial\Omega$ and their divergence misses the boundary flux (see Proposition 3.4). We now turn to the analysis of the scheme.

5. Analysis

This section establishes the stability (Section 5.1), consistency (Section 5.2) and convergence (Section 5.3) of the scheme. To alleviate the exposition, some proofs are gathered in the appendices.

5.1. **Stability.** The stability analysis relies on a discrete counterpart of the energy (2.9), called the pseudo-energy. It is built from the norms induced by the Hodge operators, which play the role of the $\boldsymbol{\varepsilon}$- and $\boldsymbol{\mu}$-weighted $\boldsymbol{L}^2$ norms in (2.9). Because of the primal/dual staggering, two families are needed: norms on primal dofs and norms on dual dofs. For $\mathsf{g} \in \mathcal{E}$, $\mathsf{b} \in \mathcal{F}$, $\boldsymbol{\phi} \in \widetilde{\mathcal{F}}$ and $\boldsymbol{\psi} \in \widetilde{\mathcal{E}}$, define

$$\|\mathsf{g}\|_\varepsilon^2 := [\![\mathsf{g}, \mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{g}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}, \qquad \|\mathsf{b}\|_{\mu^{-1}}^2 := [\![\mathsf{b}, \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\mathsf{b}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}},$$
$$\|\boldsymbol{\phi}\|_{\varepsilon^{-1}}^2 := [\![(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}, \boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}, \qquad \|\boldsymbol{\psi}\|_\mu^2 := [\![(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\boldsymbol{\psi}, \boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}.$$

Here $(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}$ is the restricted inverse of (4.1), so $\|\cdot\|_{\varepsilon^{-1}}$ is a norm on the components associated with interior edges (a seminorm on the full space $\widetilde{\mathcal{F}}$). We also write

$$\langle\!\langle \cdot, \cdot \rangle\!\rangle_\varepsilon := [\![\cdot, \mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\cdot]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} \text{ and } \langle\!\langle \cdot, \cdot \rangle\!\rangle_\mu := [\![(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\cdot, \cdot]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}. \tag{5.1}$$

For $\mathsf{g} \in \mathcal{E}_0$, $\mathsf{b} \in \mathcal{F}$, $\boldsymbol{\phi} \in \widetilde{\mathcal{F}}$ and $\boldsymbol{\psi} \in \widetilde{\mathcal{E}}$, Cauchy–Schwarz gives

$$\left|[\![\mathsf{g}, \boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}\right| \le \|\mathsf{g}\|_\varepsilon \|\boldsymbol{\phi}\|_{\varepsilon^{-1}}, \qquad \left|[\![\mathsf{b}, \boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}\right| \le \|\mathsf{b}\|_{\mu^{-1}} \|\boldsymbol{\psi}\|_\mu, \tag{5.2}$$

and the Hodge operators satisfy

$$\|(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}\|_\varepsilon = \|\boldsymbol{\phi}\|_{\varepsilon^{-1}}, \qquad \|(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\boldsymbol{\psi}\|_{\mu^{-1}} = \|\boldsymbol{\psi}\|_\mu. \tag{5.3}$$

Finally, for $(\mathsf{g}, \boldsymbol{\psi}) \in \mathcal{E} \times \widetilde{\mathcal{E}}$, set

$$\|(\mathsf{g}, \boldsymbol{\psi})\|_{\varepsilon,\mu}^2 := \|\mathsf{g}\|_\varepsilon^2 + \|\boldsymbol{\psi}\|_\mu^2. \tag{5.4}$$

The discrete counterpart of (2.9) must account for the time staggering: H is not available at t_n, so the magnetic term is replaced by the pairing of the two neighboring levels $\mathsf{H}^{n\pm\frac{1}{2}}$. Let E, H be given by the scheme (4.2)–(4.6). The pseudo-energy is the sequence $(\mathsf{W}^n)_{n\in[\![0,N]\!]}$ defined by

$$\begin{aligned}
\mathsf{W}^n &:= \frac{1}{2}\left(\|\mathsf{E}^n\|_\varepsilon^2 + \langle\!\langle \mathsf{H}^{n+\frac{1}{2}}, \mathsf{H}^{n-\frac{1}{2}} \rangle\!\rangle_\mu\right), \ \forall n \in [\![1, N-1]\!], \\
\mathsf{W}^0 &:= \frac{1}{2}\left(2\|\mathsf{E}^0\|_\varepsilon^2 + 2\langle\!\langle \mathsf{H}^0, \mathsf{H}^{\frac{1}{2}} \rangle\!\rangle_\mu + \langle\!\langle \mathsf{E}^1, \mathsf{E}^0 \rangle\!\rangle_\varepsilon\right), \\
\mathsf{W}^N &:= \frac{1}{2}\left(2\|\mathsf{E}^N\|_\varepsilon^2 + 2\langle\!\langle \mathsf{H}^N, \mathsf{H}^{N-\frac{1}{2}} \rangle\!\rangle_\mu + \langle\!\langle \mathsf{E}^N, \mathsf{E}^{N-1} \rangle\!\rangle_\varepsilon\right).
\end{aligned} \tag{5.5}$$

The endpoint formulas account for the half intervals on which E^0, H^0, E^N and H^N live. Each W^n involves three time levels. To study it as a function of the current state $(\mathsf{E}^n, \mathsf{H}^{n-\frac{1}{2}})$ only, we introduce, for any pair $(\mathsf{g}, \boldsymbol{\psi}) \in \mathcal{E}_0 \times \widetilde{\mathcal{E}}$, the quadratic form

$$\mathsf{W}_{\mathrm{s}}(\mathsf{g}, \boldsymbol{\psi}) := \frac{1}{2}\|(\mathsf{g}, \boldsymbol{\psi})\|_{\varepsilon,\mu}^2 - \frac{\Delta t}{2}[\![\mathsf{ROT}(\mathsf{g}), \boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}. \tag{5.6}$$

Lemma 5.1 (Pseudo-energy as a quadratic form)**.** With the convention $\mathsf{H}^{-\frac{1}{2}} = \mathsf{H}^0$, the pseudo-energy (5.5) satisfies

$$\mathsf{W}^n = \mathsf{W}_{\mathrm{s}}(\mathsf{E}^n, \mathsf{H}^{n-\frac{1}{2}}), \ \forall n \in [\![1, N-1]\!], \tag{5.7}$$
$$\mathsf{W}^0 = \frac{1}{2}\left(\|2\mathsf{E}^0\|_\varepsilon^2 + \|2\mathsf{H}^0\|_\mu^2 - \Delta t^2\|\mathsf{ROT}(\mathsf{E}^0)\|_{\mu^{-1}}^2\right), \tag{5.8}$$
$$\mathsf{W}^N = \mathsf{W}_{\mathrm{s}}(2\mathsf{E}^N, \mathsf{H}^{N-\frac{1}{2}}). \tag{5.9}$$

Proof. See Appendix A.1. □

A full step of the scheme maps the state $(\mathsf{E}^n, \mathsf{H}^{n-\frac{1}{2}})$ to $(\mathsf{E}^{n+1}, \mathsf{H}^{n+\frac{1}{2}})$ through (4.3); the next lemma describes the action of this map on W_{s}, for an arbitrary state.

Lemma 5.2 (Conservation by a full step)**.** Let $(\mathsf{g}, \boldsymbol{\psi}) \in \mathcal{E}_0 \times \widetilde{\mathcal{E}}$ and set

$$\boldsymbol{\psi}' := \boldsymbol{\psi} - \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\mathsf{ROT}(\mathsf{g}), \qquad \mathsf{g}' := \mathsf{g} + \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\widetilde{\mathsf{ROT}}(\boldsymbol{\psi}').$$

Then $\mathsf{W}_{\mathrm{s}}(\mathsf{g}', \boldsymbol{\psi}') = \mathsf{W}_{\mathrm{s}}(\mathsf{g}, \boldsymbol{\psi})$ for every time step Δt.

Proof. See Appendix A.2. □

Proposition 5.3 (Pseudo-energy conservation)**.** The pseudo-energy (5.5) is conserved in time, without any condition on Δt:

$$\mathsf{W}^m = \mathsf{W}^0, \ \forall m \in [\![1, N]\!]. \tag{5.10}$$

Proof. For $1 \le n \le N-2$, the update (4.3) is the step of Lemma 5.2. Thus (5.7) gives $\mathsf{W}^{n+1} = \mathsf{W}^n$. The initial and final steps are proved in Appendix A.3; together they also cover $N = 2$. □

Conservation of W^n is only useful if W_s controls the energy norm (5.4), which requires bounding the cross term of (5.6) by the quadratic part. The first ingredient is a discrete inverse inequality.

Proposition 5.4 (Inverse inequality)**.** There exists a constant $\mathsf{C}_\mathrm{inv} > 0$, depending only on the maximal number of edges of a face and of faces of a cell sharing an edge, such that

$$\|\mathsf{ROT}(\mathsf{g})\|_{2,\mathcal{F}} \le \mathsf{C}_\mathrm{inv} h_\mathrm{min}^{-1} \|\mathsf{g}\|_{2,\mathcal{E}}, \ \forall \mathsf{g} \in \mathcal{E}. \tag{5.11}$$

Proof. Apply the Cauchy–Schwarz inequality to $(\mathsf{ROT}\ \mathsf{g})_\mathrm{f}$ face by face, sum over the faces of each cell, and use $h_\mathrm{c}^{-1} \le h_\mathrm{min}^{-2} h_\mathrm{c}$; details are given in Appendix A.4. □

Combined with the Hodge bounds (**H1**), it relates $\Delta t \|\mathsf{ROT}(\mathsf{g})\|_{\mu^{-1}}$ to $\|\mathsf{g}\|_\varepsilon$ through $\Delta t / h_\mathrm{min}$, hence the need for a condition on the time step, which is the object of the next proposition.

Proposition 5.5 (CFL condition and energy norm)**.** Let $\alpha \in (0,1)$ and set

$$c_0 := (\varepsilon_\flat \mu_\flat)^{-1/2}, \qquad C_\mathrm{stab} := (\eta_\varepsilon \eta_{\mu^{-1}})^{-1/2} \mathsf{C}_\mathrm{inv}^{-1}.$$

If the CFL condition holds:

$$\Delta t \le 2 C_\mathrm{stab} \frac{h_\mathrm{min}}{c_0} (1-\alpha), \tag{5.12}$$

then the primal and dual curl bounds hold:

$$\Delta t \|\mathsf{ROT}(\mathsf{g})\|_{\mu^{-1}} \le 2(1-\alpha)\|\mathsf{g}\|_\varepsilon, \qquad \forall \mathsf{g} \in \mathcal{E}_0, \tag{5.13a}$$

$$\Delta t \|\widetilde{\mathsf{ROT}}(\boldsymbol{\psi})\|_{\varepsilon^{-1}} \le 2(1-\alpha)\|\boldsymbol{\psi}\|_\mu, \qquad \forall \boldsymbol{\psi} \in \widetilde{\mathcal{E}}. \tag{5.13b}$$

Moreover, for every $(\mathsf{g}, \boldsymbol{\psi}) \in \mathcal{E}_0 \times \widetilde{\mathcal{E}}$,

$$\frac{\alpha}{2} \|(\mathsf{g}, \boldsymbol{\psi})\|_{\varepsilon,\mu}^2 \le \mathsf{W}_\mathrm{s}(\mathsf{g}, \boldsymbol{\psi}) \le \|(\mathsf{g}, \boldsymbol{\psi})\|_{\varepsilon,\mu}^2, \tag{5.14}$$

and $\mathsf{W}_\mathrm{s}^{1/2}$ is a norm. Along the discrete trajectory, with $\mathsf{H}^{-\frac{1}{2}} = \mathsf{H}^0$, we also have

$$\mathsf{W}^n \ge \frac{\alpha}{2} \|(\mathsf{E}^n, \mathsf{H}^{n-\frac{1}{2}})\|_{\varepsilon,\mu}^2, \ \forall n \in [\![0, N]\!]. \tag{5.15}$$

Proof. See Appendix A.5. □

The condition (5.12) is a sufficient condition, explicit in terms of the mesh and material constants, but it may be pessimistic on locally refined meshes since it involves h_min. In practice, the largest time step for which (5.13a) holds on $\mathcal{E}_0$ is $2(1-\alpha)\lambda_\mathrm{max}^{-1/2}$, where λ_max is the largest eigenvalue of the generalized eigenvalue problem $\widetilde{\mathsf{ROT}}\mathsf{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{ROT}(\mathsf{g}) = \lambda \mathsf{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}} \mathsf{g}$ posed on $\mathcal{E}_0$.

Corollary 5.6 (Stability)**.** Whenever the condition (5.12) is satisfied, the numerical scheme (4.3) is energy stable:

$$\|(\mathsf{E}^n, \mathsf{H}^{n-\frac{1}{2}})\|_{\varepsilon,\mu} \le 2\alpha^{-\frac{1}{2}} \|(\mathsf{E}^0, \mathsf{H}^0)\|_{\varepsilon,\mu}, \ \forall n \in [\![0, N]\!].$$

Proof. By Proposition 5.5, Proposition 5.3 and (5.8), $\frac{\alpha}{2}\|(\mathsf{E}^n, \mathsf{H}^{n-\frac{1}{2}})\|_{\varepsilon,\mu}^2 \le \mathsf{W}^n = \mathsf{W}^0 \le 2\|(\mathsf{E}^0, \mathsf{H}^0)\|_{\varepsilon,\mu}^2$. □

5.2. **Consistency.** We estimate the discrete error in terms of the residuals in the constitutive relations. Let $(\boldsymbol{E}, \boldsymbol{H}, \boldsymbol{D}, \boldsymbol{B})$ be a sufficiently regular solution of (2.3). In the sequel, for nonnegative quantities a and b, $a \lesssim b$ means $a \le Cb$ with a constant C independent of the mesh, of Δt, of α and of the exact solution; what C depends on is made precise in each statement. We use the intervals J_n introduced in Section 3.2, and retain the endpoint convention $\mathsf{B}^{-\frac{1}{2}} = \mathsf{B}^0$, $\mathsf{B}^{N+\frac{1}{2}} = \mathsf{B}^N$.

5.2.1. *Errors and residuals.* Following Section 4, E^n and $\mathsf{H}^{n+\frac{1}{2}}$ approximate time-integrated circulations while D^n and $\mathsf{B}^{n+\frac{1}{2}}$ approximate instantaneous fluxes; the errors are defined accordingly, as the numerical unknowns minus the corresponding reductions of the exact solution. For $n \in [\![0, N]\!]$, set

$$\delta_{\boldsymbol{E}}^n := \mathsf{E}^n - \int_{J_n} \mathsf{R}_\mathcal{E}(\boldsymbol{E}(t))\, dt, \qquad \delta_{\boldsymbol{D}}^n := \mathsf{D}^n - \mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_n)).$$

With the endpoint conventions above, set

$$\delta_{\boldsymbol{B}}^{n+\frac{1}{2}} := \mathsf{B}^{n+\frac{1}{2}} - \mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_{n+\frac{1}{2}})), \qquad n \in [\![-1, N]\!],$$

and write $\delta_{\boldsymbol{B}}^{0} := \delta_{\boldsymbol{B}}^{-\frac{1}{2}}$ and $\delta_{\boldsymbol{B}}^{N} := \delta_{\boldsymbol{B}}^{N+\frac{1}{2}}$. For $n \in [\![0, N-1]\!]$ and $m \in \{0, N\}$, define

$$\delta_{\boldsymbol{H}}^{n+\frac{1}{2}} := \mathsf{H}^{n+\frac{1}{2}} - \int_{t_n}^{t_{n+1}} \mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t))\,dt, \qquad \delta_{\boldsymbol{H}}^{m} := \mathsf{H}^{m} - \int_{J_m} \mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t))\,dt.$$

The errors $\delta_{\boldsymbol{E}}^{0}, \delta_{\boldsymbol{E}}^{N}, \delta_{\boldsymbol{H}}^{0}, \delta_{\boldsymbol{H}}^{N}$ of the four endpoint averages, which are post-processed outputs of the scheme (see Appendix B.1), are not part of the analysis, whereas the final flux error $\delta_{\boldsymbol{B}}^{N}$ is estimated. The initialization $\mathsf{D}^{0} = \mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}_0)$, $\mathsf{B}^{0} = \mathsf{R}_{\mathcal{F}}(\boldsymbol{B}_0)$ gives $\delta_{\boldsymbol{D}}^{0} = \delta_{\boldsymbol{B}}^{0} = 0$, and the PEC condition implies

$$\delta_{\boldsymbol{E}}^{n} \in \mathcal{E}_0, \qquad n \in [\![0, N]\!]. \tag{5.16}$$

Indeed, $\mathsf{E}_{\mathrm{e}}^{n} = 0$ for $\mathrm{e} \in \mathrm{E}^{\partial}$ by (4.5), and $\int_{\mathrm{e}} \boldsymbol{E}(t) \cdot \boldsymbol{\tau}_{\mathrm{e}} = 0$ for $\mathrm{e} \in \mathrm{E}^{\partial}$ since $\boldsymbol{n} \times \boldsymbol{E} = 0$ on $\partial\Omega$ and a boundary edge is tangent to $\partial\Omega$; hence both E^{n} and $\int_{J_n} \mathsf{R}_{\mathcal{E}}(\boldsymbol{E})$ vanish on boundary edges, and so does their difference.

The scheme satisfies the discrete Faraday and Ampère laws exactly, so the only source of error lies in the constitutive relations: the Hodge operators, applied to time averages of the exact circulations, do not return exactly the exact fluxes. This mismatch is measured by the following residuals. Define the full-step constitutive residuals by

$$\begin{aligned}
\mathbb{E}_{\varepsilon,\Delta t}^{n} &:= \frac{1}{\Delta t}\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}} \int_{J_n} \mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t))\,dt - \mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_n)), && n \in [\![1, N-1]\!],\\
\mathbb{E}_{\mu,\Delta t}^{n+\frac{1}{2}} &:= \frac{1}{\Delta t}(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1} \int_{t_n}^{t_{n+1}} \mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t))\,dt - \mathsf{R}_{\mathcal{F}}(\boldsymbol{B}(t_{n+\frac{1}{2}})), && n \in [\![0, N-1]\!].
\end{aligned} \tag{5.17}$$

Here $\boldsymbol{D} = \boldsymbol{\varepsilon}\boldsymbol{E}$ and $\boldsymbol{B} = \boldsymbol{\mu}\boldsymbol{H}$, and the factor $1/\Delta t = |J_n|^{-1}$ turns the time integrals into averages, so that $\mathbb{E}_{\varepsilon,\Delta t}^{n}$ compares the discrete Hodge operator applied to the time average of the exact electric circulations over J_n with the exact flux of $\boldsymbol{D}$ at the instant t_n. Throughout this subsection, electric constitutive identities and Ampère's law are understood on the dual faces attached to interior edges, as in Section 4. Accordingly, $\|\!|\cdot|\!\|_{\varepsilon^{-1}}$ measures only these components.

5.2.2. *Error equations and recursion.* Integrating Maxwell's equations over the time intervals and using the commuting properties of the reduction maps gives exact balance equations. Subtracting them from the scheme, and using the definitions of the constitutive residuals, yields the full-step error system below; the derivation is given in Appendix B.1.

$$\delta_{\boldsymbol{B}}^{n+\frac{1}{2}} - \delta_{\boldsymbol{B}}^{n-\frac{1}{2}} + \mathsf{ROT}(\delta_{\boldsymbol{E}}^{n}) = 0, \quad \forall n \in [\![1, N-1]\!], \tag{5.18a}$$

$$\delta_{\boldsymbol{B}}^{n+\frac{1}{2}} = \frac{1}{\Delta t}(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\delta_{\boldsymbol{H}}^{n+\frac{1}{2}} + \mathbb{E}_{\mu,\Delta t}^{n+\frac{1}{2}}, \quad \forall n \in [\![0, N-1]\!], \tag{5.18b}$$

$$\delta_{\boldsymbol{D}}^{n+1} - \delta_{\boldsymbol{D}}^{n} - \widetilde{\mathsf{ROT}}(\delta_{\boldsymbol{H}}^{n+\frac{1}{2}}) = 0, \quad \forall n \in [\![0, N-1]\!], \tag{5.18c}$$

$$\delta_{\boldsymbol{D}}^{n+1} = \frac{1}{\Delta t}\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}}\delta_{\boldsymbol{E}}^{n+1} + \mathbb{E}_{\varepsilon,\Delta t}^{n+1}, \quad \forall n \in [\![0, N-2]\!]. \tag{5.18d}$$

The two endpoint half steps are written on the fluxes only: eliminating the endpoint electric circulations gives the flux updates

$$\mathsf{B}^{\frac{1}{2}} = \mathsf{B}^{0} - \frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{D}^{0}\big), \qquad \mathsf{B}^{N} = \mathsf{B}^{N-\frac{1}{2}} - \frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{D}^{N}\big). \tag{5.19}$$

Comparing these updates with Faraday's law integrated over J_0 and J_N defines the two half-step defects, which depend on the exact solution only:

$$\mathsf{r}^{n} := \frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_n))\big) - \mathsf{ROT}\Big(\int_{J_n} \mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t))\,dt\Big), \qquad n \in \{0, N\}. \tag{5.20}$$

Since the initial fluxes are exact, $\delta_{\boldsymbol{D}}^{0} = \delta_{\boldsymbol{B}}^{0} = 0$, the endpoint errors satisfy

$$\delta_{\boldsymbol{B}}^{\frac{1}{2}} = -\mathsf{r}^{0}, \qquad \delta_{\boldsymbol{B}}^{N} = \delta_{\boldsymbol{B}}^{N-\frac{1}{2}} - \frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\delta_{\boldsymbol{D}}^{N}\big) - \mathsf{r}^{N}, \tag{5.21}$$

where $\delta_{\boldsymbol{D}}^{N}$ is understood on the dual faces attached to interior edges. The derivation and the endpoint-output formulas are given in Appendix B.1.

Eliminating the fluxes between two consecutive constitutive relations (5.18b), resp. (5.18d), leaves a difference of consecutive residuals; the recursion for the field errors is therefore driven by the quantities

$$\mathcal{Q}_{\boldsymbol{H}}^{k-\frac{1}{2}} := \mathbb{E}_{\mu,\Delta t}^{k+\frac{1}{2}} - \mathbb{E}_{\mu,\Delta t}^{k-\frac{1}{2}}, \qquad k \in [\![1, N-1]\!], \tag{5.22}$$

$$\mathcal{Q}_{\boldsymbol{E}}^{k} := \mathbb{E}_{\varepsilon,\Delta t}^{k+1} - \mathbb{E}_{\varepsilon,\Delta t}^{k}, \qquad k \in [\![1, N-2]\!]. \tag{5.23}$$

All indices in (5.22) and (5.23) correspond to full steps.

Lemma 5.7 (Error recursion)**.** The field errors satisfy

$$\delta_{\boldsymbol{H}}^{n+\frac{1}{2}} = \delta_{\boldsymbol{H}}^{n-\frac{1}{2}} - \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{ROT}(\delta_{\boldsymbol{E}}^{n}) - \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathcal{Q}_{\boldsymbol{H}}^{n-\frac{1}{2}}, \quad \forall n \in [\![1, N-1]\!], \tag{5.24a}$$

$$\delta_{\boldsymbol{E}}^{n+1} = \delta_{\boldsymbol{E}}^{n} + \Delta t (\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1} \widetilde{\mathsf{ROT}}(\delta_{\boldsymbol{H}}^{n+\frac{1}{2}}) - \Delta t (\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1} \mathcal{Q}_{\boldsymbol{E}}^{n}, \quad \forall n \in [\![1, N-2]\!], \tag{5.24b}$$

started by

$$\delta_{\boldsymbol{H}}^{\frac{1}{2}} = -\Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \big(\mathbb{E}_{\mu,\Delta t}^{\frac{1}{2}} + \mathsf{r}^{0}\big), \qquad \delta_{\boldsymbol{E}}^{1} = \Delta t (\mathrm{H}_{\varepsilon}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1} \big(\widetilde{\mathsf{ROT}}(\delta_{\boldsymbol{H}}^{\frac{1}{2}}) - \mathbb{E}_{\varepsilon,\Delta t}^{1}\big). \tag{5.25}$$

Proof. See Appendix B.2. □

5.2.3. *Consistency estimate.* The recursion (5.24) is a homogeneous full step of the scheme, which preserves W_{s} (Lemma 5.2), plus a forcing by the residual differences. We therefore measure the error in the energy norm $\mathsf{W}_{\mathrm{s}}^{1/2}$ of Proposition 5.5, which controls $\|\!|(\cdot,\cdot)|\!\|_{\varepsilon,\mu}$ under the CFL condition (5.14). To state the estimate compactly, we collect the residuals into

$$\begin{aligned} \mathcal{R} &:= \max\Big(\max_{1\le k\le N-1} \|\!|\mathbb{E}_{\varepsilon,\Delta t}^{k}|\!\|_{\varepsilon^{-1}}, \ \max_{0\le k\le N-1} \|\!|\mathbb{E}_{\mu,\Delta t}^{k+\frac{1}{2}}|\!\|_{\mu^{-1}} \Big), \\ \mathcal{R}_{\frac{1}{2}} &:= \max\big(\|\!|\mathsf{r}^{0}|\!\|_{\mu^{-1}}, \|\!|\mathsf{r}^{N}|\!\|_{\mu^{-1}} \big). \end{aligned} \tag{5.26}$$

$\mathcal{R}$ collects the full-step constitutive residuals and $\mathcal{R}_{\frac{1}{2}}$ the two half-step defects.

Proposition 5.8 (Consistency estimate)**.** Assume the CFL condition (5.12), with empty sums equal to zero when $N = 2$. Then for all $m \in [\![1, N-1]\!]$, $n \in [\![0, N-1]\!]$ and $p \in [\![0, N]\!]$,

$$\begin{aligned} &\|\!|\delta_{\boldsymbol{E}}^{m}|\!\|_{\varepsilon} + \|\!|\delta_{\boldsymbol{H}}^{n+\frac{1}{2}}|\!\|_{\mu} + \Delta t \big(\|\!|\delta_{\boldsymbol{D}}^{p}|\!\|_{\varepsilon^{-1}} + \|\!|\delta_{\boldsymbol{B}}^{n+\frac{1}{2}}|\!\|_{\mu^{-1}} + \|\!|\delta_{\boldsymbol{B}}^{N}|\!\|_{\mu^{-1}} \big) \\ &\quad \lesssim \alpha^{-\frac{1}{2}} \Delta t \left(\mathcal{R} + \mathcal{R}_{\frac{1}{2}} + \sum_{k=1}^{N-2} \big(\|\!|\mathcal{Q}_{\boldsymbol{E}}^{k}|\!\|_{\varepsilon^{-1}} + \|\!|\mathcal{Q}_{\boldsymbol{H}}^{k-\frac{1}{2}}|\!\|_{\mu^{-1}} \big) \right). \end{aligned} \tag{5.27}$$

The hidden constant is an absolute number, independent of the mesh, Δt and α.

Proof outline. For $m \in [\![1, N-1]\!]$, set $\mathsf{X}^{m} := (\delta_{\boldsymbol{E}}^{m}, \delta_{\boldsymbol{H}}^{m-\frac{1}{2}})$, the error state before the magnetic update. We use the pair norm (5.4) and W_{s} with pairs as single arguments. The starting relations (5.25) and the CFL curl bounds give $\|\!|\mathsf{X}^{1}|\!\|_{\varepsilon,\mu} \lesssim \Delta t(\mathcal{R} + \mathcal{R}_{\frac{1}{2}})$. Each full-step error update is a homogeneous update plus a forcing pair f^{n} produced by the residual differences in (5.24). The CFL bounds imply

$$\|\!|\mathsf{f}^{n}|\!\|_{\varepsilon,\mu} \lesssim \Delta t \big(\|\!|\mathcal{Q}_{\boldsymbol{E}}^{n}|\!\|_{\varepsilon^{-1}} + \|\!|\mathcal{Q}_{\boldsymbol{H}}^{n-\frac{1}{2}}|\!\|_{\mu^{-1}} \big), \qquad n \in [\![1, N-2]\!].$$

Conservation of W_{s} by the homogeneous step and the triangle inequality yield

$$\mathsf{W}_{\mathrm{s}}(\mathsf{X}^{n+1})^{\frac{1}{2}} \le \mathsf{W}_{\mathrm{s}}(\mathsf{X}^{n})^{\frac{1}{2}} + \|\!|\mathsf{f}^{n}|\!\|_{\varepsilon,\mu}.$$

Summing over the full steps and using coercivity gives the field-error bound. The last magnetic update, the constitutive error relations and the final half step then give the flux estimates. The detailed proof is given in Appendix B.3. □

The initial half-step defect enters through the starting state, whereas the final defect contributes only to the final flux. The bounds on the constitutive residuals, their consecutive differences and the half-step defects are established in Section 5.3.

5.3. **Convergence.** We combine Proposition 5.8 with spatial and temporal residual estimates. Fix a polyhedral material partition $\mathcal{P} = \{\Omega_i\}_{i=1}^{I}$ on which $\boldsymbol{\varepsilon}$ and $\boldsymbol{\mu}$ are constant, and assume that every mesh in the family is fitted to it, *i.e.* each cell $\mathrm{c} \in \mathrm{C}$ is contained in a single Ω_i, so that the coefficients are constant on every cell. The mesh assumptions of Section 3.1 and the uniform Hodge bounds (**H1c**) are retained. Piecewise constant coefficients make the Hodge operators exact on cellwise constant fields (**H2c**), which is the property the spatial estimates rest upon.

5.3.1. *Regularity assumptions.* The residual estimates require two kinds of regularity: spatial regularity and time regularity. We introduce the corresponding spaces.

The H^1 regularity below is imposed separately in each material subdomain, rather than globally across interfaces. The material partition $\mathcal{P}$ is fixed, whereas C denotes the mesh used for local estimates.

The electric and magnetic fields are reduced by their edge circulations. We use

$$\boldsymbol{V} := \{\boldsymbol{u} \in H(\nabla\times;\Omega) : \boldsymbol{u}|_{\Omega_i} \in [H^1(\Omega_i)]^3 \ (1 \le i \le I), \ \nabla\times\boldsymbol{u} \in [L^4(\Omega)]^3\}.$$

The global $H(\nabla\times;\Omega)$ condition enforces tangential compatibility at material interfaces, while the additional curl regularity is used in the edge estimates of Lemma 5.12. For the electric field, the homogeneous tangential boundary condition is incorporated through

$$\boldsymbol{V}_0 := \boldsymbol{V} \cap H_0(\nabla\times;\Omega).$$

We equip $\boldsymbol{V}$ and its subspace $\boldsymbol{V}_0$ with the norm

$$\|\boldsymbol{u}\|_{\boldsymbol{V}}^2 := \sum_{i=1}^{I} \|\boldsymbol{u}\|_{[H^1(\Omega_i)]^3}^2 + \|\nabla\times\boldsymbol{u}\|_{[L^4(\Omega)]^3}^2.$$

The fluxes $\boldsymbol{D}=\boldsymbol{\varepsilon}\boldsymbol{E}$ and $\boldsymbol{B}=\boldsymbol{\mu}\boldsymbol{H}$ are reduced by their normal integrals over faces. Accordingly, we use

$$\boldsymbol{Y} := \{\boldsymbol{w}\in H(\nabla\cdot;\Omega) : \boldsymbol{w}|_{\Omega_i}\in[H^1(\Omega_i)]^3 \ (1\le i\le I)\}, \qquad \|\boldsymbol{w}\|_{\boldsymbol{Y}}^2 := \sum_{i=1}^{I} \|\boldsymbol{w}\|_{[H^1(\Omega_i)]^3}^2.$$

Here the global $H(\nabla\cdot;\Omega)$ condition enforces normal compatibility at material interfaces.

To bound temporal quadrature errors, we also use fields that are continuous up to the boundary of each material subdomain, without imposing global continuity:

$$\boldsymbol{C}^0_{\mathcal{P}} := \{\boldsymbol{u}:\Omega\to\mathbb{R}^3 :\ \boldsymbol{u}|_{\Omega_i}\in C^0(\overline{\Omega_i})^3 \ (1\le i\le I)\},$$
$$\|\boldsymbol{u}\|_{\boldsymbol{C}^0_{\mathcal{P}}} := \max_{1\le i\le I}\sup_{\Omega_i}|\boldsymbol{u}|.$$

This space describes spatial continuity within each material; time regularity is specified separately. The notation $C^j([0,t_F];X)$ denotes the space of j times continuously differentiable maps into the normed space X.

5.3.2. *Residual decomposition.* Each constitutive residual (5.17) mixes a time-quadrature error and the error of the Hodge operator on the exact field at a given instant; we separate these two contributions. The spatial residuals are

$$\mathbb{E}_\varepsilon(\boldsymbol{E}) := \mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}) - \mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{\varepsilon}\boldsymbol{E}),$$
$$\mathbb{E}_\mu(\boldsymbol{H}) := (\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}) - \mathsf{R}_{\mathcal{F}}(\boldsymbol{\mu}\boldsymbol{H}).$$

For $1\le n\le N-1$, the electric time residual is

$$\mathbb{E}_{\Delta t}^n(\boldsymbol{E}) := \mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\left(\frac{1}{\Delta t}\int_{J_n}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t))\,dt - \mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n))\right).$$

For $0\le n\le N-1$, the magnetic time residual is

$$\mathbb{E}_{\Delta t}^{n+1/2}(\boldsymbol{H}) := (\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\left(\frac{1}{\Delta t}\int_{t_n}^{t_{n+1}}\mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t))\,dt - \mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t_{n+1/2}))\right).$$

These residuals are those collected in $\mathcal{R}$ in (5.26); the half-step defects $\mathsf{r}^0,\mathsf{r}^N$ are treated in Lemma 5.10 below.

Adding and subtracting $\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n))$ and $(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t_{n-\frac12}))$ in (5.17) gives the decomposition into temporal and spatial parts:

$$\mathbb{E}_{\varepsilon,\Delta t}^n = \mathbb{E}_{\Delta t}^n(\boldsymbol{E}) + \mathbb{E}_\varepsilon(\boldsymbol{E}(\cdot,t_n)),\ \forall n\in[\![1,N-1]\!], \tag{5.28}$$

$$\mathbb{E}_{\mu,\Delta t}^{n-\frac12} = \mathbb{E}_{\Delta t}^{n-\frac12}(\boldsymbol{H}) + \mathbb{E}_\mu(\boldsymbol{H}(\cdot,t_{n-\frac12})),\ \forall n\in[\![1,N]\!]. \tag{5.29}$$

Proposition 5.9 (Residual decomposition)**.** Let $\boldsymbol{E}\in C^0([0,t_F];\boldsymbol{V}_0)$, $\boldsymbol{H}\in C^0([0,t_F];\boldsymbol{V})$ and $\boldsymbol{\varepsilon}\boldsymbol{E},\boldsymbol{\mu}\boldsymbol{H}\in C^0([0,t_F];\boldsymbol{Y})$. Then

$$\max_{1\le n\le N-1}\|\mathbb{E}_{\varepsilon,\Delta t}^n\|_{\varepsilon^{-1}} \le \max_{1\le n\le N-1}\|\mathbb{E}_{\Delta t}^n(\boldsymbol{E})\|_{\varepsilon^{-1}} + \sup_{t\in[0,t_F]}\|\mathbb{E}_\varepsilon(\boldsymbol{E}(t))\|_{\varepsilon^{-1}}, \tag{5.30a}$$

$$\max_{1\le n\le N}\|\mathbb{E}_{\mu,\Delta t}^{n-\frac12}\|_{\mu^{-1}} \le \max_{1\le n\le N}\|\mathbb{E}_{\Delta t}^{n-\frac12}(\boldsymbol{H})\|_{\mu^{-1}} + \sup_{t\in[0,t_F]}\|\mathbb{E}_\mu(\boldsymbol{H}(t))\|_{\mu^{-1}}. \tag{5.30b}$$

Proof. Apply the triangle inequality to (5.28) and (5.29), take the maxima over the indicated indices, and bound each sampled spatial residual by its supremum over $[0,t_F]$. □

Lemma 5.10 (Decomposition of the half-step defects)**.** Let $\boldsymbol{E} \in C^0([0,t_F];\boldsymbol{V}_0)$ with $\boldsymbol{\varepsilon E} \in C^0([0,t_F];\boldsymbol{Y})$, and $\boldsymbol{H} \in C^1([0,t_F];\boldsymbol{V})$, solving (2.3). For $n \in \{0,N\}$, the half-step defects (5.20) split as

$$\begin{aligned} \mathsf{r}^n &= -\frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathbb{E}_\varepsilon(\boldsymbol{E}(t_n))\big) + \boldsymbol{\tau}^n, \\ \boldsymbol{\tau}^n &:= \int_{J_n} \mathsf{R}_{\mathcal{F}}\big(\partial_t \boldsymbol{B}(t) - \partial_t \boldsymbol{B}(t_n)\big)\,dt, \end{aligned} \tag{5.31}$$

and, under the CFL condition (5.12),

$$\|\mathsf{r}^n\|_{\mu^{-1}} \le \|\mathbb{E}_\varepsilon(\boldsymbol{E}(t_n))\|_{\varepsilon^{-1}} + \|\boldsymbol{\tau}^n\|_{\mu^{-1}}. \tag{5.32}$$

Proof outline. Subtracting the spatial constitutive residual isolates the spatial contribution, while Faraday's law and the commuting property turn the remaining term into $\boldsymbol{\tau}^n$. The bound follows from the CFL inequality; the detailed proof is given in Appendix C.1. □

5.3.3. *Spatial residual estimates.* To estimate the spatial residual we first define local and global energy norms. For $\boldsymbol{\phi} \in \widetilde{\mathcal{F}}$ and $\boldsymbol{\psi} \in \widetilde{\mathcal{E}}$, the dual norms admit the characterizations

$$\|\boldsymbol{\phi}\|_{\varepsilon^{-1}} = \sup_{\mathsf{g}\in\mathcal{E}_0\setminus\{0\}} \frac{[\![\mathsf{g},\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}}{\|\mathsf{g}\|_\varepsilon}, \qquad \|\boldsymbol{\psi}\|_\mu = \sup_{\mathsf{b}\in\mathcal{F}\setminus\{0\}} \frac{[\![\mathsf{b},\boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}}{\|\mathsf{b}\|_{\mu^{-1}}}. \tag{5.33}$$

For each cell c, define local norms for $\mathsf{g} \in \mathcal{E}_\mathrm{c}$, $\mathsf{b} \in \mathcal{F}_\mathrm{c}$, $\boldsymbol{\phi}_\mathrm{c} \in \widetilde{\mathcal{F}}_\mathrm{c}$ and $\boldsymbol{\psi}_\mathrm{c} \in \widetilde{\mathcal{E}}_\mathrm{c}$ by

$$\begin{aligned} \|\mathsf{g}\|_{\varepsilon,\mathrm{c}}^2 &:= [\![\mathsf{g}, \mathrm{H}_\varepsilon^{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}\mathsf{g}]\!]_{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}, \qquad \|\mathsf{b}\|_{\mu^{-1},\mathrm{c}}^2 := [\![\mathsf{b}, \mathrm{H}_{\mu^{-1}}^{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}}\mathsf{b}]\!]_{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}}, \\ \|\boldsymbol{\phi}_\mathrm{c}\|_{\varepsilon^{-1},\mathrm{c}}^2 &:= [\![(\mathrm{H}_\varepsilon^{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}})^{-1}\boldsymbol{\phi}_\mathrm{c}, \boldsymbol{\phi}_\mathrm{c}]\!]_{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}, \qquad \|\boldsymbol{\psi}_\mathrm{c}\|_{\mu,\mathrm{c}}^2 := [\![(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}})^{-1}\boldsymbol{\psi}_\mathrm{c}, \boldsymbol{\psi}_\mathrm{c}]\!]_{\mathcal{F}_\mathrm{c}\widetilde{\mathcal{E}}_\mathrm{c}}. \end{aligned}$$

For global primal vectors, assembly gives, with cell restrictions understood,

$$\|\mathsf{g}\|_\varepsilon^2 = \sum_{\mathrm{c}\in\mathrm{C}} \|\mathsf{g}\|_{\varepsilon,\mathrm{c}}^2, \qquad \|\mathsf{b}\|_{\mu^{-1}}^2 = \sum_{\mathrm{c}\in\mathrm{C}} \|\mathsf{b}\|_{\mu^{-1},\mathrm{c}}^2.$$

For the dual norms, the global inverses are not the assembly of the local inverses; instead, the following inequality holds. It follows from the algebraic estimate of Codecasa and Trevisan [21, Theorem 9], also used by Bonelle and Ern [7, Lemma 3.2].

Lemma 5.11 (Assembly inequality)**.** Let $(\boldsymbol{\phi}_\mathrm{c})_{\mathrm{c}\in\mathrm{C}}$ be local dual-face vectors and let $\boldsymbol{\phi} \in \widetilde{\mathcal{F}}$ be their assembly, *i.e.* $[\![\mathsf{g},\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} = \sum_\mathrm{c} [\![\mathsf{g},\boldsymbol{\phi}_\mathrm{c}]\!]_{\mathcal{E}_\mathrm{c}\widetilde{\mathcal{F}}_\mathrm{c}}$ for all $\mathsf{g} \in \mathcal{E}_0$ (restrictions to cells are implicit). Then $\|\boldsymbol{\phi}\|_{\varepsilon^{-1}}^2 \le \sum_\mathrm{c} \|\boldsymbol{\phi}_\mathrm{c}\|_{\varepsilon^{-1},\mathrm{c}}^2$. Likewise, if $\boldsymbol{\psi} \in \widetilde{\mathcal{E}}$ is the assembly of local dual-edge vectors $(\boldsymbol{\psi}_\mathrm{c})_\mathrm{c}$, then $\|\boldsymbol{\psi}\|_\mu^2 \le \sum_\mathrm{c} \|\boldsymbol{\psi}_\mathrm{c}\|_{\mu,\mathrm{c}}^2$.

Proof. For $\mathsf{g} \in \mathcal{E}_0$, local and cellwise Cauchy–Schwarz inequalities and assembly give

$$|[\![\mathsf{g},\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}| \le \sum_\mathrm{c} \|\mathsf{g}\|_{\varepsilon,\mathrm{c}} \|\boldsymbol{\phi}_\mathrm{c}\|_{\varepsilon^{-1},\mathrm{c}} \le \|\mathsf{g}\|_\varepsilon \Big(\sum_\mathrm{c} \|\boldsymbol{\phi}_\mathrm{c}\|_{\varepsilon^{-1},\mathrm{c}}^2\Big)^{\frac{1}{2}}.$$

Taking the supremum in (5.33) proves the electric bound; the magnetic bound follows identically. □

Lemma 5.12 (Edge and face reductions of non-constant fields)**.** Let $\mathrm{c} \in \mathrm{C}$ and let π_0 be the $L^2(\mathrm{c})$-orthogonal projection on constants. Let γ be a primal edge $\mathrm{e} \in \mathrm{E}_\mathrm{c}$ or a dual-edge piece $\tilde{\mathrm{e}}_\mathrm{c}(\mathrm{f})$, with unit tangent $\boldsymbol{\tau}$, and let σ be a planar piece of a dual face $\tilde{\mathrm{f}}_\mathrm{c}(\mathrm{e})$ or a primal face $\mathrm{f} \in \mathrm{F}_\mathrm{c}$, with unit normal $\boldsymbol{n}$. For $\boldsymbol{v} \in [H^1(\mathrm{c})]^3$,

$$\Big|\int_\sigma (\boldsymbol{v} - \pi_0\boldsymbol{v})\cdot\boldsymbol{n}\Big| \lesssim h_\mathrm{c}^{\frac{3}{2}} |\boldsymbol{v}|_{[H^1(\mathrm{c})]^3}. \tag{5.34}$$

If additionally $\nabla\times\boldsymbol{v} \in [L^4(\mathrm{c})]^3$, then

$$\Big|\int_\gamma (\boldsymbol{v} - \pi_0\boldsymbol{v})\cdot\boldsymbol{\tau}\Big| \lesssim h_\mathrm{c}^{\frac{1}{2}} \big(|\boldsymbol{v}|_{[H^1(\mathrm{c})]^3} + h_\mathrm{c}^{\frac{3}{4}} \|\nabla\times\boldsymbol{v}\|_{[L^4(\mathrm{c})]^3}\big). \tag{5.35}$$

The constants depend only on the mesh regularity.

Proof outline. The face estimate follows from the trace and Poincaré inequalities. For the edge estimate, apply the extended circulation bound of Ern and Guermond [25, eq. (17.25)] to $\boldsymbol{v} - \pi_0\boldsymbol{v}$. The scaled estimates and the full argument are given in Appendix C.2. □

To collect the terms in the residual bounds, first define the broken seminorm, for fields that are H^1 on every cell, by

$$|\boldsymbol{u}|^2_{\boldsymbol{H}^1(\mathrm{C})} := \sum_{\mathrm{c}\in\mathrm{C}} |\boldsymbol{u}|^2_{[H^1(\mathrm{c})]^3}.$$

Since the mesh is fitted to $\mathcal{P}$, for fields that are H^1 on each material subdomain,

$$|\boldsymbol{u}|^2_{\boldsymbol{H}^1(\mathrm{C})} \le \sum_{i=1}^{I} |\boldsymbol{u}|^2_{[H^1(\Omega_i)]^3},$$

so this seminorm is controlled independently of the mesh. With the material weights, set

$$S_\varepsilon(\boldsymbol{u}) := \varepsilon_\sharp^{\frac12}\big(|\boldsymbol{u}|_{\boldsymbol{H}^1(\mathrm{C})} + |\Omega|^{\frac14}\|\nabla\times\boldsymbol{u}\|_{[L^4(\Omega)]^3}\big) + \varepsilon_\flat^{-\frac12}|\boldsymbol{\varepsilon u}|_{\boldsymbol{H}^1(\mathrm{C})},$$
$$S_\mu(\boldsymbol{u}) := \mu_\sharp^{\frac12}\big(|\boldsymbol{u}|_{\boldsymbol{H}^1(\mathrm{C})} + |\Omega|^{\frac14}\|\nabla\times\boldsymbol{u}\|_{[L^4(\Omega)]^3}\big) + \mu_\flat^{-\frac12}|\boldsymbol{\mu u}|_{\boldsymbol{H}^1(\mathrm{C})}.$$

For fixed fields in $\boldsymbol{V}$, the piecewise constant material tensors ensure that these quantities are bounded independently of h.

Proposition 5.13 (Spatial residuals)**.** Let $\boldsymbol{E}\in\boldsymbol{V}_0$, $\boldsymbol{H}\in\boldsymbol{V}$ and $\boldsymbol{\varepsilon E},\boldsymbol{\mu H}\in\boldsymbol{Y}$. Then

$$\|\mathbb{E}_\varepsilon(\boldsymbol{E})\|_{\varepsilon^{-1}} \lesssim h\, S_\varepsilon(\boldsymbol{E}) \quad\text{and}\quad \|\mathbb{E}_\mu(\boldsymbol{H})\|_{\mu^{-1}} \lesssim h\, S_\mu(\boldsymbol{H}).$$

The statement is about a single (time-independent) field; it is applied at each instant t to $\boldsymbol{E}(t)$, $\boldsymbol{H}(t)$, and also to the time derivatives $\partial_t\boldsymbol{E}(t)$, $\partial_t\boldsymbol{H}(t)$ in the proof of Theorem 5.15.

Proof. Write $\boldsymbol{D}=\boldsymbol{\varepsilon E}$ and $\boldsymbol{B}=\boldsymbol{\mu H}$. The lower Hodge stability bounds (3.6a) and (3.6b), by local duality and weighted Cauchy–Schwarz, give

$$\|\boldsymbol{\phi}_\mathrm{c}\|^2_{\varepsilon^{-1},\mathrm{c}} \le \eta_\varepsilon\,\varepsilon_\flat^{-1}\sum_{\mathrm{e}\in\mathrm{E_c}} h_\mathrm{c}^{-1}(\boldsymbol{\phi}_\mathrm{c})^2_{\tilde{\mathrm{f}}_\mathrm{c}(\mathrm{e})}, \tag{5.36}$$

$$\|\boldsymbol{\psi}_\mathrm{c}\|^2_{\mu,\mathrm{c}} \le \eta_{\mu^{-1}}\,\mu_\sharp\sum_{\mathrm{f}\in\mathrm{F_c}} h_\mathrm{c}(\boldsymbol{\psi}_\mathrm{c})^2_{\tilde{\mathrm{e}}_\mathrm{c}(\mathrm{f})}. \tag{5.37}$$

Electric residual. Since $\boldsymbol{\varepsilon}$ is constant on each cell, $\boldsymbol{\varepsilon}\pi_0\boldsymbol{E}=\pi_0\boldsymbol{D}$. Thus constant consistency (3.7) shows that $\mathbb{E}_\varepsilon(\boldsymbol{E})$ is the assembly of

$$\mathbb{E}_{\varepsilon,c} := \mathrm{H}_\varepsilon^{\mathcal{E}_c\widetilde{\mathcal{F}}_c}\mathsf{R}_{\mathcal{E}_c}(\boldsymbol{E}-\pi_0\boldsymbol{E}) - \mathsf{R}_{\widetilde{\mathcal{F}}_c}(\boldsymbol{D}-\pi_0\boldsymbol{D}).$$

The local Hodge isometry, the triangle inequality, and the bounds (3.6a), (5.36) and Lemma 5.12 yield

$$\begin{aligned}\|\mathbb{E}_{\varepsilon,c}\|^2_{\varepsilon^{-1},\mathrm{c}} &\le 2\|\mathsf{R}_{\mathcal{E}_c}(\boldsymbol{E}-\pi_0\boldsymbol{E})\|^2_{\varepsilon,\mathrm{c}} + 2\|\mathsf{R}_{\widetilde{\mathcal{F}}_c}(\boldsymbol{D}-\pi_0\boldsymbol{D})\|^2_{\varepsilon^{-1},\mathrm{c}}\\ &\lesssim h_\mathrm{c}^2\Big(\varepsilon_\sharp|\boldsymbol{E}|^2_{[H^1(\mathrm{c})]^3} + \varepsilon_\sharp h_\mathrm{c}^{\frac32}\|\nabla\times\boldsymbol{E}\|^2_{[L^4(\mathrm{c})]^3} + \varepsilon_\flat^{-1}|\boldsymbol{D}|^2_{[H^1(\mathrm{c})]^3}\Big).\end{aligned}$$

Here each dual face is split into planar pieces, whose number, like that of cell edges, is bounded by mesh regularity. By Cauchy–Schwarz and $\sum_\mathrm{c} h_\mathrm{c}^3\lesssim|\Omega|$,

$$\sum_\mathrm{c} h_\mathrm{c}^{\frac32}\|\nabla\times\boldsymbol{E}\|^2_{[L^4(\mathrm{c})]^3} \lesssim |\Omega|^{\frac12}\|\nabla\times\boldsymbol{E}\|^2_{[L^4(\Omega)]^3}.$$

Summing the local bound, using Lemma 5.11 and $h_\mathrm{c}\le h$, gives $\|\mathbb{E}_\varepsilon(\boldsymbol{E})\|_{\varepsilon^{-1}}\lesssim h\,S_\varepsilon(\boldsymbol{E})$.

Magnetic residual. Set $\boldsymbol{\psi} := \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\mathsf{R}_{\mathcal{F}}(\boldsymbol{B}) - \mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H})$. Then, by (5.3),

$$\mathbb{E}_\mu(\boldsymbol{H}) = -(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\boldsymbol{\psi}, \qquad \|\mathbb{E}_\mu(\boldsymbol{H})\|_{\mu^{-1}} = \|\boldsymbol{\psi}\|_\mu.$$

Since $\boldsymbol{\mu}^{-1}$ is cellwise constant, constant consistency (3.8) shows that $\boldsymbol{\psi}$ is the assembly of

$$\boldsymbol{\psi}_\mathrm{c} := \mathrm{H}_{\mu^{-1}}^{\mathcal{F}_c\widetilde{\mathcal{E}}_c}\mathsf{R}_{\mathcal{F}_c}(\boldsymbol{B}-\pi_0\boldsymbol{B}) - \mathsf{R}_{\widetilde{\mathcal{E}}_c}(\boldsymbol{H}-\pi_0\boldsymbol{H}).$$

The bounds (3.6b), (5.37) and Lemma 5.12, applied to primal faces and dual-edge pieces, similarly give

$$\|\boldsymbol{\psi}_\mathrm{c}\|^2_{\mu,\mathrm{c}} \lesssim h_\mathrm{c}^2\Big(\mu_\flat^{-1}|\boldsymbol{B}|^2_{[H^1(\mathrm{c})]^3} + \mu_\sharp|\boldsymbol{H}|^2_{[H^1(\mathrm{c})]^3} + \mu_\sharp h_\mathrm{c}^{\frac32}\|\nabla\times\boldsymbol{H}\|^2_{[L^4(\mathrm{c})]^3}\Big).$$

By Lemma 5.11 and the same summation argument, $\|\mathbb{E}_\mu(\boldsymbol{H})\|_{\mu^{-1}}\lesssim h\,S_\mu(\boldsymbol{H})$. □

5.3.4. *Temporal residual estimates.*

Proposition 5.14 (Time residuals)**.** Let $\boldsymbol{E} \in C^0([0,t_F];\boldsymbol{V}_0)$ and $\boldsymbol{H} \in C^0([0,t_F];\boldsymbol{V})$. For each available time derivative, set $\mathcal{B}_j := |\Omega|^{\frac12} \sup_{t\in[0,t_F]} \big(\varepsilon_\sharp^{\frac12} \|\partial_t^j \boldsymbol{E}(t)\|_{\boldsymbol{C}^0_{\mathcal{P}}} + \mu_\sharp^{\frac12} \|\partial_t^j \boldsymbol{H}(t)\|_{\boldsymbol{C}^0_{\mathcal{P}}}\big)$.

(i) If $\boldsymbol{E},\boldsymbol{H} \in C^2([0,t_F];\boldsymbol{C}^0_{\mathcal{P}})$, then for the full-step residuals

$$\max_{1\le n\le N-1} \|\mathbb{E}^n_{\Delta t}(\boldsymbol{E})\|_{\varepsilon^{-1}} + \max_{0\le n\le N-1} \|\mathbb{E}^{n+\frac12}_{\Delta t}(\boldsymbol{H})\|_{\mu^{-1}} \lesssim \Delta t^2\, \mathcal{B}_2. \tag{5.38}$$

(ii) If $\boldsymbol{E},\boldsymbol{H} \in C^3([0,t_F];\boldsymbol{C}^0_{\mathcal{P}})$, then the differences of consecutive full-step residuals satisfy, for all $k \in [\![1, N-2]\!]$,

$$\|\mathbb{E}^{k+1}_{\Delta t}(\boldsymbol{E}) - \mathbb{E}^k_{\Delta t}(\boldsymbol{E})\|_{\varepsilon^{-1}} + \|\mathbb{E}^{k+\frac12}_{\Delta t}(\boldsymbol{H}) - \mathbb{E}^{k-\frac12}_{\Delta t}(\boldsymbol{H})\|_{\mu^{-1}} \lesssim \Delta t^3\, \mathcal{B}_3. \tag{5.39}$$

(iii) If $\boldsymbol{H} \in C^2([0,t_F];\boldsymbol{C}^0_{\mathcal{P}})$, the time parts $\boldsymbol{\tau}^0, \boldsymbol{\tau}^N$ of the half-step defects (Lemma 5.10) satisfy

$$\|\boldsymbol{\tau}^0\|_{\mu^{-1}} + \|\boldsymbol{\tau}^N\|_{\mu^{-1}} \lesssim \Delta t^2 \Big(\frac{\mu_\sharp}{\mu_\flat}\Big)^{\frac12} \mathcal{B}_2. \tag{5.40}$$

Proof. Centered quadrature. Set $\delta := \Delta t$ and $Q_\delta f(s) := \delta^{-1} \int_{-\delta/2}^{\delta/2} f(s+r)\, dr - f(s)$. For $f \in C^2([0,t_F])$, the midpoint quadrature estimate gives

$$|Q_\delta f(s)| \le \frac{\delta^2}{24} \|f''\|_{L^\infty([0,t_F])}. \tag{5.41}$$

For $f \in C^3([0,t_F])$, the identity $(Q_\delta f)' = Q_\delta(f')$ and the mean value theorem imply

$$|Q_\delta f(s+\delta) - Q_\delta f(s)| \le \frac{\delta^3}{24} \|f'''\|_{L^\infty([0,t_F])}.$$

Here all quadrature windows are contained in $[0,t_F]$. The same estimates hold for $\boldsymbol{C}^0_{\mathcal{P}}$-valued functions, with absolute values replaced by the corresponding norms.

Full-step residuals and their differences. For $\mathsf{g} \in \mathcal{E}_0$ and $\boldsymbol{\psi} \in \widetilde{\mathcal{E}}$, the isometries (5.3) and Hodge bounds (3.10a)–(3.10b) give

$$\|\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_\varepsilon \mathsf{g}\|^2_{\varepsilon^{-1}} = \|\mathsf{g}\|^2_\varepsilon \lesssim \varepsilon_\sharp \sum_{\mathrm{c}} h_{\mathrm{c}} \sum_{\mathrm{e}\in \mathrm{E}_{\mathrm{c}}} \mathsf{g}^2_{\mathrm{e}},$$

$$\|(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}\boldsymbol{\psi}\|^2_{\mu^{-1}} = \|\boldsymbol{\psi}\|^2_\mu \lesssim \mu_\sharp \sum_{\mathrm{f}\in\mathrm{F}} \boldsymbol{\psi}^2_{\tilde{\mathrm{e}}(\mathrm{f})} \Big(\sum_{\mathrm{c}\in\mathrm{C}_{\mathrm{f}}} h_{\mathrm{c}}^{-1}\Big)^{-1}.$$

The second bound follows by weighted Cauchy–Schwarz in (5.33), using the lower magnetic Hodge bound. The time residuals are

$$\mathbb{E}^n_{\Delta t}(\boldsymbol{E}) = \mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_\varepsilon \mathsf{R}_{\mathcal{E}}\big(Q_\delta \boldsymbol{E}(t_n)\big), \qquad \mathbb{E}^{n+\frac12}_{\Delta t}(\boldsymbol{H}) = (\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1} \mathsf{R}_{\widetilde{\mathcal{E}}}\big(Q_\delta \boldsymbol{H}(t_{n+\frac12})\big).$$

The electric reductions lie in $\mathcal{E}_0$ by the PEC condition. Circulations are bounded by the edge length times the $\boldsymbol{C}^0_{\mathcal{P}}$ norm. Since $|\mathrm{e}| \le h_{\mathrm{c}}$ and $|\tilde{\mathrm{e}}(\mathrm{f})| \le \sum_{\mathrm{c}\in\mathrm{C}_{\mathrm{f}}} h_{\mathrm{c}}$, both weighted sums of squared edge lengths are $\lesssim \sum_{\mathrm{c}} h_{\mathrm{c}}^3 \lesssim |\Omega|$, by mesh regularity and bounded incidences. Thus (5.41) gives (5.38). Applying the same norm bounds to differences of $Q_\delta \boldsymbol{E}$ and $Q_\delta \boldsymbol{H}$ at consecutive centers, separated by δ, gives (5.39) using the $O(\delta^3)$ estimate above.

Half-step defects. By (5.31), the one-sided Taylor estimate on a window of length $\delta/2$ gives, for $n \in \{0, N\}$ and each face f,

$$|\boldsymbol{\tau}^n_{\mathrm{f}}| \le \frac{\delta^2}{8} |\mathrm{f}|\, \mu_\sharp \sup_{t\in[0,t_F]} \|\partial_t^2 \boldsymbol{H}(t)\|_{\boldsymbol{C}^0_{\mathcal{P}}}.$$

The upper magnetic Hodge bound (3.10b), $|\mathrm{f}|^2 h_{\mathrm{c}}^{-1} \lesssim h_{\mathrm{c}}^3$ and $\sum_{\mathrm{c}} h_{\mathrm{c}}^3 \lesssim |\Omega|$ yield

$$\|\boldsymbol{\tau}^n\|_{\mu^{-1}} \lesssim \delta^2 \mu_\flat^{-\frac12} \mu_\sharp |\Omega|^{\frac12} \sup_{t\in[0,t_F]} \|\partial_t^2 \boldsymbol{H}(t)\|_{\boldsymbol{C}^0_{\mathcal{P}}} \le \delta^2 \Big(\frac{\mu_\sharp}{\mu_\flat}\Big)^{\frac12} \mathcal{B}_2,$$

which gives (5.40). □

5.3.5. *Convergence estimate.* We now state the main result of this work.

Theorem 5.15 (Convergence)**.** Assume the CFL condition (5.12) and that the exact solution satisfies

$$\boldsymbol{E} \in C^1([0,t_F];\boldsymbol{V}_0)\cap C^3([0,t_F];\boldsymbol{C}^0_{\mathcal{P}}),\quad \boldsymbol{H} \in C^1([0,t_F];\boldsymbol{V})\cap C^3([0,t_F];\boldsymbol{C}^0_{\mathcal{P}}),$$
$$\boldsymbol{\varepsilon E},\ \boldsymbol{\mu H} \in C^1([0,t_F];\boldsymbol{Y}).$$

Let $\mathcal{A}_j := \sup_{t\in[0,t_F]}\big(S_\varepsilon(\partial_t^j \boldsymbol{E}(t)) + S_\mu(\partial_t^j \boldsymbol{H}(t))\big)$ for $j\in\{0,1\}$, let $\mathcal{B}_2,\mathcal{B}_3$ be as in Proposition 5.14, and set

$$\mathcal{M} := h\,(\mathcal{A}_0 + t_F\mathcal{A}_1) + \Delta t^2\Big(\big(1+(\mu_\sharp/\mu_\flat)^{\frac12}\big)\mathcal{B}_2 + t_F\mathcal{B}_3\Big).$$

For all $m\in[\![1,N-1]\!]$, $n\in[\![0,N-1]\!]$ and $p\in[\![0,N]\!]$,

$$\|\Delta t^{-1}\delta^m_{\boldsymbol{E}}\|_\varepsilon + \|\Delta t^{-1}\delta^{n+\frac12}_{\boldsymbol{H}}\|_\mu + \|\delta^p_{\boldsymbol{D}}\|_{\varepsilon^{-1}} + \|\delta^{n+\frac12}_{\boldsymbol{B}}\|_{\mu^{-1}} + \|\delta^N_{\boldsymbol{B}}\|_{\mu^{-1}} \lesssim \alpha^{-\frac12}\,\mathcal{M}. \tag{5.42}$$

The hidden constants depend only on the mesh regularity and the uniform Hodge stability constants.

Proof. The residual decomposition (Proposition 5.9 and Lemma 5.10) and the spatial and temporal estimates (Propositions 5.13 and 5.14) give

$$\mathcal{R} \lesssim h\mathcal{A}_0 + \Delta t^2\mathcal{B}_2, \qquad \mathcal{R}_{\frac12} \lesssim h\mathcal{A}_0 + \Delta t^2\Big(\frac{\mu_\sharp}{\mu_\flat}\Big)^{\frac12}\mathcal{B}_2.$$

For $1\le k\le N-2$, the decomposition (5.28) and the linearity and continuity of the spatial residual operator (Proposition 5.13) yield

$$\mathcal{Q}^k_{\boldsymbol{E}} = \mathbb{E}^{k+1}_{\Delta t}(\boldsymbol{E}) - \mathbb{E}^k_{\Delta t}(\boldsymbol{E}) + \int_{t_k}^{t_{k+1}} \mathbb{E}_\varepsilon(\partial_t \boldsymbol{E}(t))\,dt.$$

The same argument applies to $\mathcal{Q}^{k-\frac12}_{\boldsymbol{H}}$, which involves only the full-step magnetic residuals at $t_{k-\frac12}$ and $t_{k+\frac12}$. Thus (5.39) and Proposition 5.13, applied to the time derivatives, give

$$\|\mathcal{Q}^k_{\boldsymbol{E}}\|_{\varepsilon^{-1}} + \|\mathcal{Q}^{k-\frac12}_{\boldsymbol{H}}\|_{\mu^{-1}} \lesssim \Delta t\, h\mathcal{A}_1 + \Delta t^3\mathcal{B}_3.$$

Summing and using $(N-2)\Delta t\le t_F$, with an empty sum when $N=2$, gives

$$\sum_{k=1}^{N-2}\big(\|\mathcal{Q}^k_{\boldsymbol{E}}\|_{\varepsilon^{-1}} + \|\mathcal{Q}^{k-\frac12}_{\boldsymbol{H}}\|_{\mu^{-1}}\big) \lesssim t_F(h\mathcal{A}_1 + \Delta t^2\mathcal{B}_3).$$

Inserting these bounds into the consistency estimate (5.27) and dividing by Δt proves (5.42) with $\mathcal{M}$ as defined above. □

5.3.6. *Reconstructed-field convergence.* To end this section we translate Theorem 5.15 into L^2 estimates using stable reconstructions that are exact on constants. These additional assumptions do not enter the algebraic analysis above. Following the reconstruction framework of [7, Sections 4.2 and 5.2.1], let $\mathrm{L}_{\mathcal{E}_\mathrm{c}} : \mathcal{E}_\mathrm{c} \to [L^2(\mathrm{c})]^3$ and $\mathrm{L}_{\mathcal{F}_\mathrm{c}} : \mathcal{F}_\mathrm{c} \to [L^2(\mathrm{c})]^3$ be linear maps such that

(**R1**) For a constant C_L independent of the cell and the mesh,

$$\|\mathrm{L}_{\mathcal{E}_\mathrm{c}}(\mathsf{g})\|_{[L^2(\mathrm{c})]^3} \le C_L\|\mathsf{g}\|_{\mathcal{E},\mathrm{c}}, \qquad \|\mathrm{L}_{\mathcal{F}_\mathrm{c}}(\mathsf{b})\|_{[L^2(\mathrm{c})]^3} \le C_L\|\mathsf{b}\|_{\mathcal{F},\mathrm{c}}. \tag{5.43}$$

(**R2**) For every $\boldsymbol{K}\in[\mathbb{P}_0(\mathrm{c})]^3$,

$$\mathrm{L}_{\mathcal{E}_\mathrm{c}}\mathsf{R}_{\mathcal{E}_\mathrm{c}}(\boldsymbol{K}) = \boldsymbol{K}, \qquad \mathrm{L}_{\mathcal{F}_\mathrm{c}}\mathsf{R}_{\mathcal{F}_\mathrm{c}}(\boldsymbol{K}) = \boldsymbol{K}. \tag{5.44}$$

The global reconstructions are defined cellwise, for $\mathsf{X}\in\{\mathcal{E},\mathcal{F}\}$, by

$$\mathrm{L}_\mathsf{X}(\mathsf{u})|_\mathrm{c} := \mathrm{L}_{\mathsf{X}_\mathrm{c}}\big(\mathsf{P}_{\mathsf{X}_\mathrm{c}}\mathsf{u}\big), \tag{5.45}$$

where $\mathsf{P}_{\mathsf{X}_\mathrm{c}}$ is the restriction map of Section 3.4 from global vectors to the degrees of freedom attached to cell c.

With $\boldsymbol{V}_\mathrm{c}$ and $\boldsymbol{Y}_\mathrm{c}$ denoting the restrictions of the global regularity spaces to c, composing reconstruction with reduction defines the local approximation maps

$$\mathsf{A}_{\mathcal{E}_\mathrm{c}} := \mathrm{L}_{\mathcal{E}_\mathrm{c}}\mathsf{R}_{\mathcal{E}_\mathrm{c}} : \boldsymbol{V}_\mathrm{c} \to [L^2(\mathrm{c})]^3 \ \text{ and } \ \mathsf{A}_{\mathcal{F}_\mathrm{c}} := \mathrm{L}_{\mathcal{F}_\mathrm{c}}\mathsf{R}_{\mathcal{F}_\mathrm{c}} : \boldsymbol{Y}_\mathrm{c} \to [L^2(\mathrm{c})]^3.$$

The corresponding global approximation maps are

$$\mathsf{A}_\mathcal{E} := \mathrm{L}_\mathcal{E}\mathsf{R}_\mathcal{E} : \boldsymbol{V} \to [L^2(\Omega)]^3 \ \text{ and } \ \mathsf{A}_\mathcal{F} := \mathrm{L}_\mathcal{F}\mathsf{R}_\mathcal{F} : \boldsymbol{Y} \to [L^2(\Omega)]^3. \tag{5.46}$$

By (5.45), $(\mathsf{A}_\mathcal{E}\boldsymbol{u})|_\mathrm{c} = \mathsf{A}_{\mathcal{E}_\mathrm{c}}(\boldsymbol{u}|_\mathrm{c})$ for $\boldsymbol{u}\in\boldsymbol{V}$, and similarly for $\mathsf{A}_\mathcal{F}$ on $\boldsymbol{Y}$.

Lemma 5.16 (Spatial approximation)**.** Assume (**R1**)–(**R2**). For $\boldsymbol{u} \in \boldsymbol{V}$ and $\boldsymbol{w} \in \boldsymbol{Y}$,

$$\|\boldsymbol{u} - \mathsf{A}_{\mathcal{E}}\boldsymbol{u}\|_{[L^2(\Omega)]^3} \lesssim h\big(|\boldsymbol{u}|_{\boldsymbol{H}^1(\mathrm{C})} + |\Omega|^{1/4}\|\nabla \times \boldsymbol{u}\|_{[L^4(\Omega)]^3}\big), \tag{5.47}$$

$$\|\boldsymbol{w} - \mathsf{A}_{\mathcal{F}}\boldsymbol{w}\|_{[L^2(\Omega)]^3} \lesssim h|\boldsymbol{w}|_{\boldsymbol{H}^1(\mathrm{C})}. \tag{5.48}$$

The constants depend only on the mesh regularity and C_L.

Proof. As in the proof of [7, Proposition 4.3], subtract the cellwise mean $\pi_0\boldsymbol{u}$ (resp. $\pi_0\boldsymbol{w}$). By (**R2**), Poincaré's inequality, (**R1**) and Lemma 5.12,

$$\|\boldsymbol{u} - \mathsf{A}_{\mathcal{E}_\mathrm{c}}\boldsymbol{u}\|_{[L^2(\mathrm{c})]^3} \lesssim h_\mathrm{c}\big(|\boldsymbol{u}|_{[H^1(\mathrm{c})]^3} + h_\mathrm{c}^{3/4}\|\nabla \times \boldsymbol{u}\|_{[L^4(\mathrm{c})]^3}\big),$$

and

$$\|\boldsymbol{w} - \mathsf{A}_{\mathcal{F}_\mathrm{c}}\boldsymbol{w}\|_{[L^2(\mathrm{c})]^3} \lesssim h_\mathrm{c}|\boldsymbol{w}|_{[H^1(\mathrm{c})]^3}.$$

Sum the squared bounds, use $h_\mathrm{c} \le h$, and apply Cauchy–Schwarz and mesh regularity to obtain

$$\sum_{\mathrm{c}} h_\mathrm{c}^{3/2}\|\nabla \times \boldsymbol{u}\|^2_{[L^4(\mathrm{c})]^3} \lesssim |\Omega|^{1/2}\|\nabla \times \boldsymbol{u}\|^2_{[L^4(\Omega)]^3}.$$

This proves (5.47) and (5.48). □

The circulation unknowns are time integrals. Their normalized versions and the fluxes are

$$\begin{aligned} E_h^n &:= |J_n|^{-1}\mathsf{E}^n, \quad D_h^n := \mathsf{D}^n \quad (0 \le n \le N),\\ H_h^{n+1/2} &:= \Delta t^{-1}\mathsf{H}^{n+1/2}, \quad B_h^{n+1/2} := \mathsf{B}^{n+1/2} \quad (0 \le n \le N-1),\\ H_h^q &:= |J_q|^{-1}\mathsf{H}^q, \quad B_h^q := \mathsf{B}^q \quad (q \in \{0, N\}). \end{aligned} \tag{5.49}$$

Following the constitutive reconstructions of Bonelle and Ern [7, Corollaries 4.4 and 5.5], we define the dual reconstructions from the primal ones:

$$\mathrm{L}_{\widetilde{\mathcal{F}}}(\mathsf{d}) := \boldsymbol{\varepsilon}\mathrm{L}_{\mathcal{E}}\big((\mathrm{H}_{\boldsymbol{\varepsilon}}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{d}\big), \qquad \mathrm{L}_{\widetilde{\mathcal{E}}}(\mathsf{h}) := \boldsymbol{\mu}^{-1}\mathrm{L}_{\mathcal{F}}\big((\mathrm{H}_{\boldsymbol{\mu}^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\mathsf{h}\big). \tag{5.50}$$

The electric inverse is the restricted global inverse of (4.1): only components attached to interior edges are used.

Corollary 5.17 (Reconstructed error estimates)**.** Assume (**R1**)–(**R2**) and all the hypotheses of Theorem 5.15. Let $\mathcal{M}$ be defined there and set $\mathcal{T}_B := \{0, \frac{1}{2}, \frac{3}{2}, \dots, N - \frac{1}{2}, N\}$, with $t_0 = 0$ and $t_N = t_F$. Then

$$\begin{aligned} &\max_{0\le n\le N}\Big(\|\boldsymbol{E}(t_n) - \mathrm{L}_{\mathcal{E}}(E_h^n)\|_{[L^2(\Omega)]^3} + \|\boldsymbol{D}(t_n) - \mathrm{L}_{\widetilde{\mathcal{F}}}(D_h^n)\|_{[L^2(\Omega)]^3}\Big)\\ &\quad + \max_{r\in\mathcal{T}_B}\Big(\|\boldsymbol{B}(t_r) - \mathrm{L}_{\mathcal{F}}(B_h^r)\|_{[L^2(\Omega)]^3} + \|\boldsymbol{H}(t_r) - \mathrm{L}_{\widetilde{\mathcal{E}}}(H_h^r)\|_{[L^2(\Omega)]^3}\Big) \lesssim \alpha^{-1/2}\mathcal{M}. \end{aligned} \tag{5.51}$$

The constant depends only on the mesh regularity, C_L, the uniform Hodge stability constants and the material bounds $\varepsilon_\flat, \varepsilon_\sharp, \mu_\flat, \mu_\sharp$. In particular, for fixed material and regularity bounds and fixed α, all four reconstructed fields converge in $\boldsymbol{L}^2(\Omega)$ at order $h + \Delta t^2$ at their respective time levels.

Proof. By (**R1**), the lower Hodge stability bounds, Lemma 5.16, Theorem 5.15 and the definition of $\mathcal{A}_0$,

$$\|\boldsymbol{B}(t_r) - \mathrm{L}_{\mathcal{F}}(B_h^r)\|_{[L^2(\Omega)]^3} \lesssim \|\delta_{\boldsymbol{B}}^r\|_{\mu^{-1}} + h|\boldsymbol{B}(t_r)|_{\boldsymbol{H}^1(\mathrm{C})} \lesssim \alpha^{-1/2}\mathcal{M}.$$

For $r = 0$, use $\delta_{\boldsymbol{B}}^0 = 0$. The scheme gives $E_h^n = (\mathrm{H}_{\boldsymbol{\varepsilon}}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}D_h^n$ at every electric time level, including the endpoints. Hence

$$E_h^n - \mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n)) = (\mathrm{H}_{\boldsymbol{\varepsilon}}^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\big(\delta_{\boldsymbol{D}}^n - \mathbb{E}_\varepsilon(\boldsymbol{E}(t_n))\big).$$

Using (5.3), (**R1**), the lower electric Hodge bound, Lemma 5.16, Proposition 5.13 and Theorem 5.15, we obtain

$$\|\boldsymbol{E}(t_n) - \mathrm{L}_{\mathcal{E}}(E_h^n)\|_{[L^2(\Omega)]^3} \lesssim \|\delta_{\boldsymbol{D}}^n\|_{\varepsilon^{-1}} + hS_\varepsilon(\boldsymbol{E}(t_n)) \lesssim \alpha^{-1/2}\mathcal{M}.$$

Finally, the discrete constitutive relations and (5.50) give

$$\mathrm{L}_{\widetilde{\mathcal{F}}}(D_h^n) = \boldsymbol{\varepsilon}\mathrm{L}_{\mathcal{E}}(E_h^n), \qquad \mathrm{L}_{\widetilde{\mathcal{E}}}(H_h^r) = \boldsymbol{\mu}^{-1}\mathrm{L}_{\mathcal{F}}(B_h^r).$$

Together with the exact constitutive relations and the material bounds, these identities give the same estimate for $\boldsymbol{D}$ and $\boldsymbol{H}$. □

In particular, the endpoint outputs $E_h^0, E_h^N, H_h^0, H_h^N$ are included: being compared with the instantaneous fields $\boldsymbol{E}(t_q)$, $\boldsymbol{H}(t_q)$, they inherit the accuracy of the fluxes D^q, B^q they are computed from. Their errors with respect to the half-interval averages, left aside in Section 5.2, are not involved here.

6. Conclusion and perspectives

We have analyzed a Compatible Discrete Operator scheme for Maxwell's equations in the time domain on polyhedral meshes. The scheme rests on the separation, inherited from the geometric structure of the laws of electromagnetism, between topological relations and constitutive relations. The former are discretized exactly, in space by the incidence matrices of a primal and a dual mesh and in time by the fundamental theorem of calculus on a primal and a dual time mesh; the latter are discretized by cellwise assembled discrete Hodge operators, which carry all the metric and material information and all the approximation error. This separation was used throughout the analysis. The exactness of the discrete differential operators yields the exact conservation of the discrete divergences and of the boundary magnetic flux. The adjoint character of the primal and dual curl operators yields the exact conservation of a discrete pseudo-energy for any time step. Finally, the commuting properties of the reduction maps concentrate the consistency error in the Hodge operators, in space, and in the time quadrature of the constitutive relations, in time, which leads to an error estimate of order $h + \Delta t^2$ in the discrete energy norms for the fluxes and the full-interval time averages, under the sole algebraic requirements of symmetry, uniform stability and exactness on constant fields for the local Hodge operators. The estimate holds for anisotropic material tensors that are piecewise constant on a partition fitted by the mesh, for exact fields that are only piecewise H^1, under the additional regularity assumptions of Theorem 5.15. The hidden constant in the discrete estimate depends only on the mesh regularity and on the Hodge stability constants. Under additional stability and constant-exactness assumptions on the reconstruction operators, these discrete estimates also yield convergence of all four reconstructed fields in $\boldsymbol{L}^2(\Omega)$, with the same order $h + \Delta t^2$ at their respective time levels.

Several directions are left open. The first one concerns the boundary conditions: only the perfectly electrically conducting condition has been considered, whereas scattering problems require absorbing conditions of Silver–Müller type and the injection of incident fields. The second one concerns the Hodge operators themselves. The analysis only uses their algebraic properties, so it applies verbatim to *hybrid* choices, for instance diagonal operators on the Cartesian cells, where the scheme reduces to the Yee scheme, and dense operators on the polyhedral cells near the objects; quantifying the influence of this choice on the sharp CFL condition and on the computational cost is of practical interest. The third one concerns the construction of higher-order Hodge operators within the same topological framework. It is a natural way to improve the spatial accuracy while preserving the structural properties established here.

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Appendix A. Proofs of the stability results

This appendix contains the proofs used in Section 5.1. Electric states belong to $\mathcal{E}_0$. The trajectory identities concern the scheme (4.2)–(4.6).

A.1. **Proof of Lemma 5.1.** We repeatedly use two identities which follow from (5.1), the symmetry of the Hodge operators, the definition (4.1) and the adjunction property (3.5b): for all $\mathsf{g} \in \mathcal{E}_0$, $\boldsymbol{\psi} \in \widetilde{\mathcal{E}}$,

$$\langle\!\langle \boldsymbol{\psi}, \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{ROT}(\mathsf{g}) \rangle\!\rangle_\mu = [\![\mathsf{ROT}(\mathsf{g}), \boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}, \qquad \langle\!\langle \mathsf{g}, (\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1} \widetilde{\mathsf{ROT}}(\boldsymbol{\psi}) \rangle\!\rangle_\varepsilon = [\![\mathsf{ROT}(\mathsf{g}), \boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}. \tag{A.1}$$

(i) Let $n \in [\![1, N-1]\!]$. By (4.3a)–(4.3b), $\mathsf{H}^{n+\frac12} = \mathsf{H}^{n-\frac12} - \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{ROT}(\mathsf{E}^n)$. Inserting this in (5.5) and using the first identity of (A.1) gives (5.7).

(ii) By (4.2), (4.6) and (4.3) at $n = 0$,

$$\mathsf{H}^{\frac12} = 2\mathsf{H}^0 - \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{ROT}(\mathsf{E}^0) \ \text{ and } \ \mathsf{E}^1 = 2\mathsf{E}^0 + \Delta t (\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1} \widetilde{\mathsf{ROT}}(\mathsf{H}^{\frac12}).$$

Hence, by (A.1), $2\langle\!\langle \mathsf{H}^0, \mathsf{H}^{\frac12} \rangle\!\rangle_\mu = \|2\mathsf{H}^0\|_\mu^2 - 2\Delta t [\![\mathsf{ROT}(\mathsf{E}^0), \mathsf{H}^0]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}$ and

$$\langle\!\langle \mathsf{E}^1, \mathsf{E}^0 \rangle\!\rangle_\varepsilon = 2\|\mathsf{E}^0\|_\varepsilon^2 + \Delta t [\![\mathsf{ROT}(\mathsf{E}^0), \mathsf{H}^{\frac12}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} = 2\|\mathsf{E}^0\|_\varepsilon^2 + 2\Delta t [\![\mathsf{ROT}(\mathsf{E}^0), \mathsf{H}^0]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} - \Delta t^2 \|\mathsf{ROT}(\mathsf{E}^0)\|_{\mu^{-1}}^2.$$

Summing these two relations with $2\|\mathsf{E}^0\|_\varepsilon^2$ gives (5.8).

(iii) By (4.4), (4.3c) at $n = N-1$ and (4.3d) at $n = N-2$,

$$2\mathsf{H}^N = \mathsf{H}^{N-\frac12} - \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}} \mathsf{ROT}(\mathsf{E}^N) \ \text{ and } \ \mathsf{E}^{N-1} = 2\mathsf{E}^N - \Delta t (\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1} \widetilde{\mathsf{ROT}}(\mathsf{H}^{N-\frac12}).$$

Hence, by (A.1),

$$2\langle\!\langle \mathsf{H}^N, \mathsf{H}^{N-\frac12} \rangle\!\rangle_\mu = \|\mathsf{H}^{N-\frac12}\|_\mu^2 - \Delta t [\![\mathsf{ROT}(\mathsf{E}^N), \mathsf{H}^{N-\frac12}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} \ \text{ and } \ \langle\!\langle \mathsf{E}^N, \mathsf{E}^{N-1} \rangle\!\rangle_\varepsilon = 2\|\mathsf{E}^N\|_\varepsilon^2 - \Delta t [\![\mathsf{ROT}(\mathsf{E}^N), \mathsf{H}^{N-\frac12}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}.$$

Summing these two relations with $2\|\mathsf{E}^N\|_\varepsilon^2$ gives $2\mathsf{W}^N = \|2\mathsf{E}^N\|_\varepsilon^2 + \|\mathsf{H}^{N-\frac12}\|_\mu^2 - 2\Delta t [\![\mathsf{ROT}(\mathsf{E}^N), \mathsf{H}^{N-\frac12}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}$, which is $2\mathsf{W}_{\mathrm{s}}(2\mathsf{E}^N, \mathsf{H}^{N-\frac12})$ by (5.6), i.e. (5.9).

A.2. **Proof of Lemma 5.2.** By the update formulas and adjointness (3.5b),

$$\|\mathsf{g}'\|_\varepsilon^2 - \|\mathsf{g}\|_\varepsilon^2 = \Delta t[\![\mathsf{ROT}(\mathsf{g}'+\mathsf{g}),\boldsymbol{\psi}']\!]_{\mathcal{F}\widetilde{\mathcal{E}}},$$
$$\|\boldsymbol{\psi}'\|_\mu^2 - \|\boldsymbol{\psi}\|_\mu^2 = -\Delta t[\![\mathsf{ROT}(\mathsf{g}),\boldsymbol{\psi}'+\boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}.$$

Adding these identities gives

$$\|(\mathsf{g}',\boldsymbol{\psi}')\|_{\varepsilon,\mu}^2 - \|(\mathsf{g},\boldsymbol{\psi})\|_{\varepsilon,\mu}^2 = \Delta t\big([\![\mathsf{ROT}(\mathsf{g}'),\boldsymbol{\psi}']\!]_{\mathcal{F}\widetilde{\mathcal{E}}} - [\![\mathsf{ROT}(\mathsf{g}),\boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}\big).$$

Subtracting the cross terms in (5.6) proves the result.

A.3. **Initial and final steps of Proposition 5.3.** *(i) Initial step.* We prove $\mathsf{W}^1 = \mathsf{W}^0$. Let $\boldsymbol{\phi} := \widetilde{\mathsf{ROT}}(\mathsf{H}^{\frac12})$ and recall the updates $\mathsf{H}^{\frac12} = 2\mathsf{H}^0 - \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\mathsf{ROT}(\mathsf{E}^0)$ and $\mathsf{E}^1 = 2\mathsf{E}^0 + \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}$ from part *(ii)* of the proof of Lemma 5.1. By (5.7) with $n=1$ and (3.5b),

$$2\mathsf{W}^1 = \|\mathsf{E}^1\|_\varepsilon^2 + \|\mathsf{H}^{\frac12}\|_\mu^2 - \Delta t[\![\mathsf{E}^1,\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}.$$

Since $\|(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}\|_\varepsilon^2 = [\![(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi},\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} = \|\boldsymbol{\phi}\|_{\varepsilon^{-1}}^2$, substituting E^1 gives

$$\|\mathsf{E}^1\|_\varepsilon^2 - \Delta t[\![\mathsf{E}^1,\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} = 4\|\mathsf{E}^0\|_\varepsilon^2 + 2\Delta t[\![\mathsf{E}^0,\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} = 4\|\mathsf{E}^0\|_\varepsilon^2 + 2\Delta t[\![\mathsf{ROT}(\mathsf{E}^0),\mathsf{H}^{\frac12}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}.$$

Substituting now $\mathsf{H}^{\frac12}$ and using (A.1), we get

$$\|\mathsf{H}^{\frac12}\|_\mu^2 = 4\|\mathsf{H}^0\|_\mu^2 - 4\Delta t[\![\mathsf{ROT}(\mathsf{E}^0),\mathsf{H}^0]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} + \Delta t^2\|\mathsf{ROT}(\mathsf{E}^0)\|_{\mu^{-1}}^2,$$
$$2\Delta t[\![\mathsf{ROT}(\mathsf{E}^0),\mathsf{H}^{\frac12}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} = 4\Delta t[\![\mathsf{ROT}(\mathsf{E}^0),\mathsf{H}^0]\!]_{\mathcal{F}\widetilde{\mathcal{E}}} - 2\Delta t^2\|\mathsf{ROT}(\mathsf{E}^0)\|_{\mu^{-1}}^2.$$

Adding the three last identities yields $2\mathsf{W}^1 = \|2\mathsf{E}^0\|_\varepsilon^2 + \|2\mathsf{H}^0\|_\mu^2 - \Delta t^2\|\mathsf{ROT}(\mathsf{E}^0)\|_{\mu^{-1}}^2$, which is $2\mathsf{W}^0$ by (5.8).

(ii) Final step. We prove $\mathsf{W}^N = \mathsf{W}^{N-1}$. Let now $\boldsymbol{\phi} := \widetilde{\mathsf{ROT}}(\mathsf{H}^{N-\frac12})$. By part *(iii)* of the proof of Lemma 5.1, $\mathsf{E}^{N-1} = 2\mathsf{E}^N - \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}$, and (4.3a)–(4.3b) at $n = N-1$ give $\mathsf{H}^{N-\frac32} = \mathsf{H}^{N-\frac12} + \Delta t \mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\mathsf{ROT}(\mathsf{E}^{N-1})$. Inserting the latter in the definition (5.5) of W^{N-1} and using (A.1) and (3.5b),

$$2\mathsf{W}^{N-1} = \|\mathsf{E}^{N-1}\|_\varepsilon^2 + \|\mathsf{H}^{N-\frac12}\|_\mu^2 + \Delta t[\![\mathsf{E}^{N-1},\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}.$$

Substituting E^{N-1},

$$\|\mathsf{E}^{N-1}\|_\varepsilon^2 + \Delta t[\![\mathsf{E}^{N-1},\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} = 4\|\mathsf{E}^N\|_\varepsilon^2 - 2\Delta t[\![\mathsf{E}^N,\boldsymbol{\phi}]\!]_{\mathcal{E}\widetilde{\mathcal{F}}} = 4\|\mathsf{E}^N\|_\varepsilon^2 - 2\Delta t[\![\mathsf{ROT}(\mathsf{E}^N),\mathsf{H}^{N-\frac12}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}},$$

so that $2\mathsf{W}^{N-1}$ coincides with $2\mathsf{W}^N = 2\mathsf{W}_{\mathrm{s}}(2\mathsf{E}^N,\mathsf{H}^{N-\frac12})$ of (5.9), expanded with (5.6).

A.4. **Inverse inequality** (5.11)**.** Let $\mathrm{c}\in\mathrm{C}$ and $\mathrm{f}\in\mathrm{F_c}$. Since $\iota_{\mathrm{e,f}} = \pm1$, the Cauchy–Schwarz inequality gives

$$(\mathsf{ROT}\ \mathsf{g})_{\mathrm{f}}^2 = \left(\sum_{\mathrm{e}\in\mathrm{E_f}} \iota_{\mathrm{e,f}}\mathsf{g}_{\mathrm{e}}\right)^2 \le \mathtt{C}_1 \sum_{\mathrm{e}\in\mathrm{E_f}} \mathsf{g}_{\mathrm{e}}^2, \quad \text{with}\ \ \mathtt{C}_1 = \max_{\mathrm{f}\in\mathrm{F}}(\#\mathrm{E_f}).$$

Summing over $\mathrm{f}\in\mathrm{F_c}$ and exchanging the sums, each edge $\mathrm{e}\in\mathrm{E_c}$ is counted $\#(\mathrm{F_c}\cap\mathrm{F_e})$ times, whence

$$\sum_{\mathrm{f}\in\mathrm{F_c}} (\mathsf{ROT}\ \mathsf{g})_{\mathrm{f}}^2 \le \mathtt{C}_1\mathtt{C}_2 \sum_{\mathrm{e}\in\mathrm{E_c}} \mathsf{g}_{\mathrm{e}}^2, \quad \text{with}\ \ \mathtt{C}_2 = \max_{\mathrm{c}\in\mathrm{C}}\max_{\mathrm{e}\in\mathrm{E_c}} \#(\mathrm{F_c}\cap\mathrm{F_e}).$$

Multiplying by $h_{\mathrm{c}}^{-1} \le h_{\min}^{-2}h_{\mathrm{c}}$ and summing over the cells gives $\|\mathsf{ROT}(\mathsf{g})\|_{2,\mathcal{F}}^2 \le \mathtt{C}_1\mathtt{C}_2 h_{\min}^{-2}\|\mathsf{g}\|_{2,\mathcal{E}}^2$, which is (5.11) with $\mathtt{C}_{\mathrm{inv}} = (\mathtt{C}_1\mathtt{C}_2)^{\frac12}$.

A.5. **Proof of Proposition 5.5.** *(i) Curl bounds.* For $\mathsf{g}\in\mathcal{E}_0$, the Hodge bounds (3.10b), (3.10a) and the inverse inequality (5.11) give

$$\|\mathsf{ROT}(\mathsf{g})\|_{\mu^{-1}} \le (\eta_\varepsilon\eta_{\mu^{-1}})^{1/2}\mathtt{C}_{\mathrm{inv}}(\varepsilon_\flat\mu_\flat)^{-1/2}h_{\min}^{-1}\|\mathsf{g}\|_\varepsilon = \frac{c_0}{C_{\mathrm{stab}}h_{\min}}\|\mathsf{g}\|_\varepsilon.$$

The CFL condition yields (5.13a). Adjointness and the discrete Cauchy–Schwarz inequality imply

$$|[\![\mathsf{g},\widetilde{\mathsf{ROT}}(\boldsymbol{\psi})]\!]_{\mathcal{E}\widetilde{\mathcal{F}}}| = |[\![\mathsf{ROT}(\mathsf{g}),\boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}| \le 2(1-\alpha)\Delta t^{-1}\|\mathsf{g}\|_\varepsilon\|\boldsymbol{\psi}\|_\mu.$$

Taking the supremum over $\mathsf{g}\in\mathcal{E}_0$ with $\|\mathsf{g}\|_\varepsilon = 1$ gives the dual bound in (5.13).

(ii) Energy norm. By (5.13a) and Young's inequality,

$$\Delta t|[\![\mathsf{ROT}(\mathsf{g}),\boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}}| \le 2(1-\alpha)\|\mathsf{g}\|_\varepsilon\|\boldsymbol{\psi}\|_\mu \le (1-\alpha)\|(\mathsf{g},\boldsymbol{\psi})\|_{\varepsilon,\mu}^2.$$

Inserting this in (5.6) proves (5.14). Thus W_{s} is a positive definite quadratic form and its square root is a norm. By (5.7), this also gives (5.15) for $1\le n\le N-1$.

(iii) Initial step. For $n=0$, squaring (5.13a) and using $(1-\alpha)^2 \le 1-\alpha$ give $\Delta t^2 \|\mathsf{ROT}(\mathsf{E}^0)\|_{\mu^{-1}}^2 \le 4(1-\alpha)\|\mathsf{E}^0\|_\varepsilon^2$. Hence, by (5.8),

$$\mathsf{W}^0 \ge \frac{1}{2}\left(4\alpha\|\mathsf{E}^0\|_\varepsilon^2 + 4\|\mathsf{H}^0\|_\mu^2\right) \ge 2\alpha\|(\mathsf{E}^0,\mathsf{H}^0)\|_{\varepsilon,\mu}^2 \ge \frac{\alpha}{2}\|(\mathsf{E}^0,\mathsf{H}^0)\|_{\varepsilon,\mu}^2.$$

(iv) Final step. For $n=N$, since $2\mathsf{E}^N \in \mathcal{E}_0$, (5.9) and (5.14) give

$$\mathsf{W}^N = \mathsf{W}_{\rm s}(2\mathsf{E}^N, \mathsf{H}^{N-\frac12}) \ge \frac{\alpha}{2}\|(2\mathsf{E}^N, \mathsf{H}^{N-\frac12})\|_{\varepsilon,\mu}^2 \ge \frac{\alpha}{2}\|(\mathsf{E}^N, \mathsf{H}^{N-\frac12})\|_{\varepsilon,\mu}^2.$$

Appendix B. Proofs of the consistency results

B.1. **Derivation of the error equations. Error equations.** The exact fields satisfy $\partial_t \boldsymbol{B} = -\nabla\times\boldsymbol{E}$ and $\partial_t \boldsymbol{D} = \nabla\times\boldsymbol{H}$. By the commuting property of Proposition 3.2, $\mathsf{R}_\mathcal{F}(\nabla\times\boldsymbol{E}(t)) = \mathsf{ROT}(\mathsf{R}_\mathcal{E}(\boldsymbol{E}(t)))$, and $\mathsf{R}_\mathcal{F}$ commutes with the time derivative; integrating Faraday's law over $J_n = [t_{n-\frac12}, t_{n+\frac12}]$ gives

$$\mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_{n+\frac12})) - \mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_{n-\frac12})) + \mathsf{ROT}\Big(\int_{J_n} \mathsf{R}_\mathcal{E}(\boldsymbol{E}(t))\,dt\Big) = 0, \qquad n \in [\![0,N]\!], \tag{B.1}$$

with $t_{-\frac12} = t_0$ and $t_{N+\frac12} = t_N$ (half intervals for $n \in \{0,N\}$). Likewise, by the partial commuting property of Proposition 3.4, $\mathsf{R}_{\widetilde{\mathcal{F}}}(\nabla\times\boldsymbol{H}(t))$ and $\widetilde{\mathsf{ROT}}(\mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t)))$ coincide on the dual faces attached to interior edges, and integrating Ampère's law over $[t_n, t_{n+1}]$ gives $\mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_{n+1})) - \mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_n)) - \widetilde{\mathsf{ROT}}\big(\int_{t_n}^{t_{n+1}} \mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t))\,dt\big) = 0$ on these dual faces. Subtracting these two identities from (4.3a) and (4.3c) gives the topological error equations (5.18a) and (5.18c). For the constitutive ones, (4.3b) reads $\mathsf{B}^{n+\frac12} = \frac{1}{\Delta t}(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\mathsf{H}^{n+\frac12}$, so that, by the definitions of $\delta_{\boldsymbol{H}}^{n+\frac12}$ and $\mathbb{E}_{\mu,\Delta t}^{n+\frac12}$,

$$\begin{aligned}
\delta_{\boldsymbol{B}}^{n+\frac12} &= \frac{1}{\Delta t}(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\mathsf{H}^{n+\frac12} - \mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_{n+\frac12})) \\
&= \frac{1}{\Delta t}(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\delta_{\boldsymbol{H}}^{n+\frac12} + \Big(\frac{1}{\Delta t}(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\int_{t_n}^{t_{n+1}} \mathsf{R}_{\widetilde{\mathcal{E}}}(\boldsymbol{H}(t))\,dt - \mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_{n+\frac12}))\Big) \\
&= \frac{1}{\Delta t}(\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}})^{-1}\delta_{\boldsymbol{H}}^{n+\frac12} + \mathbb{E}_{\mu,\Delta t}^{n+\frac12};
\end{aligned}$$

and by (4.3d) and the definition (4.1), $\mathsf{D}^{n+1} = \frac{1}{\Delta t}\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{E}^{n+1}$ on the dual faces attached to interior edges, so that the same computation with $\delta_{\boldsymbol{E}}^{n+1}$ and $\mathbb{E}_{\varepsilon,\Delta t}^{n+1}$ ($n+1 \le N-1$) gives (5.18d).

Half steps. The two half steps ($n=0$ and $n=N$ in (4.3a)) are written on the fluxes. Inserting $\mathsf{E}^0 = \frac{\Delta t}{2}(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{D}^0$ from (4.2) into (4.3a) at $n=0$ (with $\mathsf{B}^{-\frac12} = \mathsf{B}^0$), and $\mathsf{E}^N = \frac{\Delta t}{2}(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{D}^N$ from (4.4) into $\mathsf{B}^N = \mathsf{B}^{N-\frac12} - \mathsf{ROT}(\mathsf{E}^N)$, gives (5.19). The factor $\frac{\Delta t}{2}$ is the length of the half interval J_0 (or J_N): $\frac{\Delta t}{2}(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{D}^0$ is the rectangle-rule approximation of $\int_{J_0}\mathsf{R}_\mathcal{E}(\boldsymbol{E})$ from the value at t_0. Hence the fluxes D^n ($0 \le n \le N$), $\mathsf{B}^{n+\frac12}$ ($0 \le n \le N-1$), B^N and the full-interval averages E^n ($1 \le n \le N-1$), $\mathsf{H}^{n+\frac12}$ ($0 \le n \le N-1$) are determined by $(\mathsf{D}^0, \mathsf{B}^0)$ through (5.19) and the full-step updates (4.3), while the four endpoint averages

$$\begin{aligned}
\mathsf{E}^0 &= \frac{\Delta t}{2}(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{D}^0, \qquad \mathsf{H}^0 = \frac{\Delta t}{2}\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\mathsf{B}^0, \\
\mathsf{E}^N &= \frac{\Delta t}{2}(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{D}^N, \qquad \mathsf{H}^N = \frac{\Delta t}{2}\mathrm{H}_{\mu^{-1}}^{\mathcal{F}\widetilde{\mathcal{E}}}\mathsf{B}^N
\end{aligned} \tag{B.2}$$

(this is (4.2), (4.6) and (4.4)) are outputs that enter no other update. The error made by the two rectangle rules is isolated in two vectors. Namely, the exact solution satisfies the half-interval Faraday identities (B.1) for $n \in \{0,N\}$,

$$\mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_{\frac12})) = \mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_0)) - \mathsf{ROT}\Big(\int_{J_0}\mathsf{R}_\mathcal{E}(\boldsymbol{E})\Big), \qquad \mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_N)) = \mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_{N-\frac12})) - \mathsf{ROT}\Big(\int_{J_N}\mathsf{R}_\mathcal{E}(\boldsymbol{E})\Big).$$

The resulting half-step defects are defined in (5.20). Subtracting the exact identities from (5.19) and using the linearity of ROT and $(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}$,

$$\begin{aligned}
\delta_{\boldsymbol{B}}^{\frac12} &= \big(\mathsf{B}^0 - \mathsf{R}_\mathcal{F}(\boldsymbol{B}(t_0))\big) - \frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}(\mathsf{D}^0 - \mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_0)))\big) \\
&\quad - \Big[\frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_0))\big) - \mathsf{ROT}\Big(\int_{J_0}\mathsf{R}_\mathcal{E}(\boldsymbol{E})\Big)\Big] = \delta_{\boldsymbol{B}}^0 - \frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\delta_{\boldsymbol{D}}^0\big) - \mathsf{r}^0,
\end{aligned}$$

and the same computation holds at t_N. Since $\delta^0_{\boldsymbol{B}}=0$ and $\delta^0_{\boldsymbol{D}}=0$, the two half steps give (5.21), where $\delta^N_{\boldsymbol{D}}$ is understood on the dual faces attached to interior edges.

B.2. **Proof of Lemma 5.7.** *(i) Magnetic recursion* (5.24a). Let $n\in[\![1,N-1]\!]$. Since $n\geq 1$, the constitutive error equation (5.18b) is available both at the index n and at the index $n-1\geq 0$:

$$\delta^{n+\frac12}_{\boldsymbol{B}}=\frac{1}{\Delta t}(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}\delta^{n+\frac12}_{\boldsymbol{H}}+\mathbb{E}^{n+\frac12}_{\mu,\Delta t},\qquad \delta^{n-\frac12}_{\boldsymbol{B}}=\frac{1}{\Delta t}(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}\delta^{n-\frac12}_{\boldsymbol{H}}+\mathbb{E}^{n-\frac12}_{\mu,\Delta t}.$$

Inserting both in (5.18a), $\delta^{n+\frac12}_{\boldsymbol{B}}-\delta^{n-\frac12}_{\boldsymbol{B}}+\mathsf{ROT}(\delta^n_{\boldsymbol{E}})=0$, gives

$$\frac{1}{\Delta t}(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}(\delta^{n+\frac12}_{\boldsymbol{H}}-\delta^{n-\frac12}_{\boldsymbol{H}})=-\mathsf{ROT}(\delta^n_{\boldsymbol{E}})-\big(\mathbb{E}^{n+\frac12}_{\mu,\Delta t}-\mathbb{E}^{n-\frac12}_{\mu,\Delta t}\big),$$

and the bracket is $\mathcal{Q}^{n-\frac12}_{\boldsymbol{H}}$ by the definition (5.22). Multiplying by $\Delta t\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}$ (so that $\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}=\mathrm{Id}$ on $\widetilde{\mathcal{E}}$) gives (5.24a).

(ii) Electric recursion (5.24b). Let $n\in[\![1,N-2]\!]$. The constitutive error equation (5.18d), $\delta^{k+1}_{\boldsymbol{D}}=\frac{1}{\Delta t}\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon}\delta^{k+1}_{\boldsymbol{E}}+\mathbb{E}^{k+1}_{\varepsilon,\Delta t}$, is available at the indices $k=n$ and $k=n-1\geq 0$, that is for $\delta^{n+1}_{\boldsymbol{D}}$ and $\delta^n_{\boldsymbol{D}}$. Inserting both in (5.18c), $\delta^{n+1}_{\boldsymbol{D}}-\delta^n_{\boldsymbol{D}}-\widetilde{\mathsf{ROT}}(\delta^{n+\frac12}_{\boldsymbol{H}})=0$, gives

$$\frac{1}{\Delta t}\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon}(\delta^{n+1}_{\boldsymbol{E}}-\delta^n_{\boldsymbol{E}})+\big(\mathbb{E}^{n+1}_{\varepsilon,\Delta t}-\mathbb{E}^n_{\varepsilon,\Delta t}\big)=\widetilde{\mathsf{ROT}}(\delta^{n+\frac12}_{\boldsymbol{H}}),$$

where the bracket is $\mathcal{Q}^n_{\boldsymbol{E}}$ by (5.23). All terms are vectors of $\widetilde{\mathcal{F}}$ and the identity holds on the dual faces attached to interior edges. Apply $\Delta t(\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon})^{-1}$: since $\delta^{n+1}_{\boldsymbol{E}}-\delta^n_{\boldsymbol{E}}\in\mathcal{E}_0$ by (5.16), $(\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon})^{-1}\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon}(\delta^{n+1}_{\boldsymbol{E}}-\delta^n_{\boldsymbol{E}})=\delta^{n+1}_{\boldsymbol{E}}-\delta^n_{\boldsymbol{E}}$ by uniqueness in (4.1), and by linearity we obtain (5.24b).

(iii) The start (5.25). By (5.18b) at $n=0$, $\delta^{\frac12}_{\boldsymbol{B}}=\frac{1}{\Delta t}(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}\delta^{\frac12}_{\boldsymbol{H}}+\mathbb{E}^{\frac12}_{\mu,\Delta t}$, and by (5.21), $\delta^{\frac12}_{\boldsymbol{B}}=-\mathsf{r}^0$. Hence $\frac{1}{\Delta t}(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}\delta^{\frac12}_{\boldsymbol{H}}=-\mathsf{r}^0-\mathbb{E}^{\frac12}_{\mu,\Delta t}$; multiplying by $\Delta t\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}$ gives the first relation. By (5.18c) at $n=0$ with $\delta^0_{\boldsymbol{D}}=0$, $\delta^1_{\boldsymbol{D}}=\widetilde{\mathsf{ROT}}(\delta^{\frac12}_{\boldsymbol{H}})$, and by (5.18d) at $n=0$ (allowed since $N\geq 2$), $\delta^1_{\boldsymbol{D}}=\frac{1}{\Delta t}\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon}\delta^1_{\boldsymbol{E}}+\mathbb{E}^1_{\varepsilon,\Delta t}$. Hence $\frac{1}{\Delta t}\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon}\delta^1_{\boldsymbol{E}}=\widetilde{\mathsf{ROT}}(\delta^{\frac12}_{\boldsymbol{H}})-\mathbb{E}^1_{\varepsilon,\Delta t}$ on the dual faces attached to interior edges; applying $\Delta t(\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon})^{-1}$ as in (ii) (with $\delta^1_{\boldsymbol{E}}\in\mathcal{E}_0$) gives the second relation.

B.3. **Proof of Proposition 5.8.** The recursion (5.24) is one homogeneous full step of the scheme plus a forcing. We therefore use the energy norm $\mathsf{W}_{\mathrm{s}}^{1/2}$ of Proposition 5.5, which is preserved by each homogeneous full step (Lemma 5.2). Recall that: for all $(\mathsf{g},\boldsymbol{\psi})\in\mathcal{E}_0\times\widetilde{\mathcal{E}}$, by (5.6) and (5.14),

$$\mathsf{W}_{\mathrm{s}}(\mathsf{g},\boldsymbol{\psi})=\frac12\|(\mathsf{g},\boldsymbol{\psi})\|^2_{\varepsilon,\mu}-\frac{\Delta t}{2}[\![\mathsf{ROT}(\mathsf{g}),\boldsymbol{\psi}]\!]_{\mathcal{F}\widetilde{\mathcal{E}}},\qquad \frac{\alpha}{2}\|(\mathsf{g},\boldsymbol{\psi})\|^2_{\varepsilon,\mu}\leq\mathsf{W}_{\mathrm{s}}(\mathsf{g},\boldsymbol{\psi})\leq\|(\mathsf{g},\boldsymbol{\psi})\|^2_{\varepsilon,\mu},$$

$\mathsf{W}_{\mathrm{s}}^{1/2}$ is a norm, and, with $\boldsymbol{\psi}':=\boldsymbol{\psi}-\Delta t\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}\mathsf{ROT}(\mathsf{g})$ and $\mathsf{g}':=\mathsf{g}+\Delta t(\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon})^{-1}\widetilde{\mathsf{ROT}}(\boldsymbol{\psi}')$, $\mathsf{W}_{\mathrm{s}}(\mathsf{g}',\boldsymbol{\psi}')=\mathsf{W}_{\mathrm{s}}(\mathsf{g},\boldsymbol{\psi})$.

First, we introduce the following notations. Set $q^k_E:=\|\mathcal{Q}^k_{\boldsymbol{E}}\|_{\varepsilon^{-1}}$, $q^k_H:=\|\mathcal{Q}^{k-\frac12}_{\boldsymbol{H}}\|_{\mu^{-1}}$ and $\Sigma:=\Delta t\sum_{k=1}^{N-2}(q^k_E+q^k_H)$, so that the right-hand side of (5.27) is $\alpha^{-\frac12}(\Delta t\,\mathcal{R}+\Delta t\,\mathcal{R}_{\frac12}+\Sigma)$; $\Sigma=0$ when $N=2$. We use the error states X^m introduced in the proof outline of Proposition 5.8.

Second, we bound the residuals. For $1\leq k\leq N-2$, $\mathcal{Q}^k_{\boldsymbol{E}}=\mathbb{E}^{k+1}_{\varepsilon,\Delta t}-\mathbb{E}^k_{\varepsilon,\Delta t}$ with both indices in $[\![1,N-1]\!]$, and for $1\leq k\leq N-1$, $\mathcal{Q}^{k-\frac12}_{\boldsymbol{H}}=\mathbb{E}^{k+\frac12}_{\mu,\Delta t}-\mathbb{E}^{k-\frac12}_{\mu,\Delta t}$ with both indices in $\{\frac12,\dots,N-\frac12\}$. By the triangle inequality and the definition (5.26) of $\mathcal{R}$, $q^k_E,q^k_H\leq 2\mathcal{R}$ for $1\leq k\leq N-2$, and $q^{N-1}_H\leq 2\mathcal{R}$. Also $\|\mathbb{E}^{\frac12}_{\mu,\Delta t}\|_{\mu^{-1}},\|\mathbb{E}^1_{\varepsilon,\Delta t}\|_{\varepsilon^{-1}}\leq\mathcal{R}$ and $\|\mathsf{r}^0\|_{\mu^{-1}},\|\mathsf{r}^N\|_{\mu^{-1}}\leq\mathcal{R}_{\frac12}$.

Third, to bound the operators we repeatedly use the isometries (5.3) together with the two CFL bounds (5.13) and $1-\alpha\leq 1$, which give, for all $\mathsf{g}\in\mathcal{E}_0$ and $\boldsymbol{\psi}\in\widetilde{\mathcal{E}}$,

$$\begin{aligned}\Delta t\|\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}\mathsf{ROT}(\mathsf{g})\|_{\mu}&=\Delta t\|\mathsf{ROT}(\mathsf{g})\|_{\mu^{-1}}\leq 2\|\mathsf{g}\|_{\varepsilon},\\ \Delta t\|(\mathrm{H}^{\mathcal{E}\widetilde{\mathcal{F}}}_{\varepsilon})^{-1}\widetilde{\mathsf{ROT}}(\boldsymbol{\psi})\|_{\varepsilon}&=\Delta t\|\widetilde{\mathsf{ROT}}(\boldsymbol{\psi})\|_{\varepsilon^{-1}}\leq 2\|\boldsymbol{\psi}\|_{\mu}.\end{aligned}\tag{B.3}$$

The second line is used three times below: in (i) for $\delta^1_{\boldsymbol{E}}$, in (ii) for the electric forcing f^n_E, and in (iii) for $\delta^N_{\boldsymbol{D}}$.

(i) Initial step. By (5.25), $\delta^{\frac12}_{\boldsymbol{H}}=-\Delta t\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}(\mathbb{E}^{\frac12}_{\mu,\Delta t}+\mathsf{r}^0)$, so the isometry $\|\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}\mathsf{b}\|_{\mu}=\|\mathsf{b}\|_{\mu^{-1}}$ and the triangle inequality give

$$\|\delta^{\frac12}_{\boldsymbol{H}}\|_{\mu}=\Delta t\|\mathbb{E}^{\frac12}_{\mu,\Delta t}+\mathsf{r}^0\|_{\mu^{-1}}\leq\Delta t\,\mathcal{R}+\Delta t\,\mathcal{R}_{\frac12}.$$

By (5.25), $\delta^1_{\boldsymbol{E}} = \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}(\widetilde{\mathsf{ROT}}(\delta^{\frac12}_{\boldsymbol{H}}) - \mathbb{L}^1_{\varepsilon,\Delta t})$, so the isometry $\|(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}\|_\varepsilon = \|\boldsymbol{\phi}\|_{\varepsilon^{-1}}$, the triangle inequality, (B.3) and $\|\mathbb{L}^1_{\varepsilon,\Delta t}\|_{\varepsilon^{-1}} \le \mathcal{R}$ give

$$\|\delta^1_{\boldsymbol{E}}\|_\varepsilon \le \Delta t\|\widetilde{\mathsf{ROT}}(\delta^{\frac12}_{\boldsymbol{H}})\|_{\varepsilon^{-1}} + \Delta t\|\mathbb{L}^1_{\varepsilon,\Delta t}\|_{\varepsilon^{-1}} \le 2\|\delta^{\frac12}_{\boldsymbol{H}}\|_\mu + \Delta t\,\mathcal{R} \le 3\Delta t\,\mathcal{R} + 2\Delta t\,\mathcal{R}_{\frac12}.$$

Hence

$$\|\mathsf{X}^1\|_{\varepsilon,\mu} \le \|\delta^1_{\boldsymbol{E}}\|_\varepsilon + \|\delta^{\frac12}_{\boldsymbol{H}}\|_\mu \le 4\Delta t\,\mathcal{R} + 3\Delta t\,\mathcal{R}_{\frac12} \le 4\Delta t\big(\mathcal{R} + \mathcal{R}_{\frac12}\big). \tag{B.4}$$

(ii) Full steps. Let $n \in [\![1, N-2]\!]$ and $(\mathsf{g}, \boldsymbol{\psi}) := \mathsf{X}^n = (\delta^n_{\boldsymbol{E}}, \delta^{n-\frac12}_{\boldsymbol{H}})$, with $\mathsf{g} \in \mathcal{E}_0$. With the notation of Lemma 5.2, $\boldsymbol{\psi}' = \boldsymbol{\psi} - \Delta t\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}\mathsf{ROT}(\mathsf{g})$ and $\mathsf{g}' = \mathsf{g} + \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\widetilde{\mathsf{ROT}}(\boldsymbol{\psi}')$, the relation (5.24a) reads $\delta^{n+\frac12}_{\boldsymbol{H}} = \boldsymbol{\psi}' + \mathsf{f}^n_H$ with $\mathsf{f}^n_H := -\Delta t\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}}\mathcal{Q}^{n-\frac12}_{\boldsymbol{H}}$, and (5.24b), by linearity of $\widetilde{\mathsf{ROT}}$ and $(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}$,

$$\delta^{n+1}_{\boldsymbol{E}} = \mathsf{g} + \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\widetilde{\mathsf{ROT}}(\boldsymbol{\psi}') + \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\widetilde{\mathsf{ROT}}(\mathsf{f}^n_H) - \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathcal{Q}^n_{\boldsymbol{E}} = \mathsf{g}' + \mathsf{f}^n_E,$$

with $\mathsf{f}^n_E := \Delta t(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\big(\widetilde{\mathsf{ROT}}(\mathsf{f}^n_H) - \mathcal{Q}^n_{\boldsymbol{E}}\big)$. In short,

$$\mathsf{X}^{n+1} = (\mathsf{g}', \boldsymbol{\psi}') + \mathsf{f}^n, \qquad \mathsf{f}^n := (\mathsf{f}^n_E, \mathsf{f}^n_H).$$

The forcing is small: $\|\mathsf{f}^n_H\|_\mu = \Delta t\|\mathcal{Q}^{n-\frac12}_{\boldsymbol{H}}\|_{\mu^{-1}} = \Delta t q^n_H$ and, by (B.3), $\|\mathsf{f}^n_E\|_\varepsilon \le 2\|\mathsf{f}^n_H\|_\mu + \Delta t\|\mathcal{Q}^n_{\boldsymbol{E}}\|_{\varepsilon^{-1}} = 2\Delta t q^n_H + \Delta t q^n_E$, so that

$$\|\mathsf{f}^n\|_{\varepsilon,\mu} \le \|\mathsf{f}^n_E\|_\varepsilon + \|\mathsf{f}^n_H\|_\mu \le 3\Delta t q^n_H + \Delta t q^n_E \le 3\Delta t(q^n_E + q^n_H).$$

Since $\mathsf{W}_{\mathrm{s}}^{1/2}$ is a norm (by Proposition 5.5, so that the triangle inequality holds), is conserved by the homogeneous step (Lemma 5.2, applicable since $\mathsf{g} \in \mathcal{E}_0$), and is bounded by $\|\cdot\|_{\varepsilon,\mu}$ (upper bound in (5.14)),

$$\mathsf{W}_{\mathrm{s}}(\mathsf{X}^{n+1})^{\frac12} \le \mathsf{W}_{\mathrm{s}}(\mathsf{g}', \boldsymbol{\psi}')^{\frac12} + \mathsf{W}_{\mathrm{s}}(\mathsf{f}^n)^{\frac12} \le \mathsf{W}_{\mathrm{s}}(\mathsf{X}^n)^{\frac12} + \|\mathsf{f}^n\|_{\varepsilon,\mu} \le \mathsf{W}_{\mathrm{s}}(\mathsf{X}^n)^{\frac12} + 3\Delta t\,(q^n_E + q^n_H).$$

Summing from $n = 1$ to $n = m-1$ for $m \in [\![2, N-1]\!]$ (the intermediate terms $\mathsf{W}_{\mathrm{s}}(\mathsf{X}^n)^{1/2}$ cancel telescopically), and using (5.14) at both ends (lower bound at X^m, upper bound at X^1) together with (B.4),

$$\Big(\frac{\alpha}{2}\Big)^{\frac12}\|\mathsf{X}^m\|_{\varepsilon,\mu} \le \mathsf{W}_{\mathrm{s}}(\mathsf{X}^m)^{\frac12} \le \|\mathsf{X}^1\|_{\varepsilon,\mu} + 3\Delta t\sum_{k=1}^{m-1}(q^k_E + q^k_H) \le 4\Delta t\big(\mathcal{R} + \mathcal{R}_{\frac12}\big) + 3\,\Sigma,$$

that is, for all $m \in [\![1, N-1]\!]$ (the case $m = 1$ being (B.4) with $(2/\alpha)^{1/2} \ge 1$),

$$\|\mathsf{X}^m\|_{\varepsilon,\mu} \le \Big(\frac{2}{\alpha}\Big)^{\frac12}\big(4\Delta t\,\mathcal{R} + 4\Delta t\,\mathcal{R}_{\frac12} + 3\,\Sigma\big). \tag{B.5}$$

This bounds $\|\delta^m_{\boldsymbol{E}}\|_\varepsilon$ for $m \in [\![1, N-1]\!]$ and $\|\delta^{n+\frac12}_{\boldsymbol{H}}\|_\mu$ for $n \in [\![0, N-2]\!]$.

(iii) Fluxes and final half step. Write $Y := \|\mathsf{X}^{N-1}\|_{\varepsilon,\mu}$, bounded by (B.5).

The last magnetic unknown. By (5.24a) with $n = N-1$, the triangle inequality, (B.3) and $q^{N-1}_H \le 2\mathcal{R}$,

$$\|\delta^{N-\frac12}_{\boldsymbol{H}}\|_\mu \le \|\delta^{N-\frac32}_{\boldsymbol{H}}\|_\mu + 2\|\delta^{N-1}_{\boldsymbol{E}}\|_\varepsilon + \Delta t q^{N-1}_H \le 3Y + 2\Delta t\,\mathcal{R}.$$

Magnetic fluxes. By (5.18b) and the isometry $\|(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}\boldsymbol{\psi}\|_{\mu^{-1}} = \|\boldsymbol{\psi}\|_\mu$, for $n \in [\![0, N-1]\!]$, $\Delta t\|\delta^{n+\frac12}_{\boldsymbol{B}}\|_{\mu^{-1}} \le \|(\mathrm{H}^{\mathcal{F}\widetilde{\mathcal{E}}}_{\mu^{-1}})^{-1}\delta^{n+\frac12}_{\boldsymbol{H}}\|_{\mu^{-1}} + \Delta t\|\mathbb{E}^{n+\frac12}_{\mu,\Delta t}\|_{\mu^{-1}} \le \|\delta^{n+\frac12}_{\boldsymbol{H}}\|_\mu + \Delta t\,\mathcal{R}$; in particular $\Delta t\|\delta^{N-\frac12}_{\boldsymbol{B}}\|_{\mu^{-1}} \le 3Y + 3\Delta t\,\mathcal{R}$.

Electric fluxes. We have $\delta^0_{\boldsymbol{D}} = 0$. By (5.18d) and the isometry $\|\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{g}\|_{\varepsilon^{-1}} = \|\mathsf{g}\|_\varepsilon$, for $p \in [\![1, N-1]\!]$, $\Delta t\|\delta^p_{\boldsymbol{D}}\|_{\varepsilon^{-1}} \le \|\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\delta^p_{\boldsymbol{E}}\|_{\varepsilon^{-1}} + \Delta t\|\mathbb{E}^p_{\varepsilon,\Delta t}\|_{\varepsilon^{-1}} \le \|\delta^p_{\boldsymbol{E}}\|_\varepsilon + \Delta t\,\mathcal{R}$. By (5.18c) with $n = N-1$ and (B.3),

$$\Delta t\|\delta^N_{\boldsymbol{D}}\|_{\varepsilon^{-1}} \le \Delta t\|\delta^{N-1}_{\boldsymbol{D}}\|_{\varepsilon^{-1}} + \Delta t\|\widetilde{\mathsf{ROT}}(\delta^{N-\frac12}_{\boldsymbol{H}})\|_{\varepsilon^{-1}} \le \Delta t\|\delta^{N-1}_{\boldsymbol{D}}\|_{\varepsilon^{-1}} + 2\|\delta^{N-\frac12}_{\boldsymbol{H}}\|_\mu \le 7Y + 5\Delta t\,\mathcal{R}.$$

Final magnetic flux. By (5.21), $\delta^N_{\boldsymbol{B}} = \delta^{N-\frac12}_{\boldsymbol{B}} - \frac{\Delta t}{2}\mathsf{ROT}((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\delta^N_{\boldsymbol{D}}) - \mathsf{r}^N$. By the primal bound in (B.3) applied to $\mathsf{g} = (\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\delta^N_{\boldsymbol{D}} \in \mathcal{E}_0$ and the isometry $\|(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}\|_\varepsilon = \|\boldsymbol{\phi}\|_{\varepsilon^{-1}}$, $\Delta t\|\frac{\Delta t}{2}\mathsf{ROT}((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\delta^N_{\boldsymbol{D}})\|_{\mu^{-1}} \le \Delta t\|(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\delta^N_{\boldsymbol{D}}\|_\varepsilon = \Delta t\|\delta^N_{\boldsymbol{D}}\|_{\varepsilon^{-1}}$. Hence, multiplying (5.21) by Δt and using the triangle inequality,

$$\begin{aligned}\Delta t\|\delta^N_{\boldsymbol{B}}\|_{\mu^{-1}} &\le \Delta t\|\delta^{N-\frac12}_{\boldsymbol{B}}\|_{\mu^{-1}} + \Delta t\|\delta^N_{\boldsymbol{D}}\|_{\varepsilon^{-1}} + \Delta t\|\mathsf{r}^N\|_{\mu^{-1}}\\ &\le (3Y + 3\Delta t\,\mathcal{R}) + (7Y + 5\Delta t\,\mathcal{R}) + \Delta t\,\mathcal{R}_{\frac12} = 10Y + 8\Delta t\,\mathcal{R} + \Delta t\,\mathcal{R}_{\frac12}.\end{aligned}$$

All the quantities of (5.27) are thus bounded by a fixed multiple of $\|\mathsf{X}^{N-1}\|_{\varepsilon,\mu} + \Delta t\,\mathcal{R} + \Delta t\,\mathcal{R}_{\frac12}$ (or of $\|\mathsf{X}^{n+1}\|_{\varepsilon,\mu} + \Delta t\,\mathcal{R}$ for $n \le N-2$), and (B.5) gives (5.27) (recall $\alpha < 1$, so that $\Delta t\,\mathcal{R} \le \alpha^{-1/2}\Delta t\,\mathcal{R}$).

Appendix C. Proofs of the convergence results

C.1. **Proof of Lemma 5.10.** Recall the definition (5.20),

$$\mathsf{r}^n = \frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_n))\big) - \mathsf{ROT}\Big(\int_{J_n}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t))\,dt\Big).$$

(i) Spatial part. Since $\boldsymbol{E}(t_n)\in\boldsymbol{V}_0$, $\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n))\in\mathcal{E}_0$ and $(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n)) = \mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n))$ by uniqueness in (4.1). Adding and subtracting $\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n))$ inside $(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}$ and using linearity,

$$(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathsf{R}_{\widetilde{\mathcal{F}}}(\boldsymbol{D}(t_n)) = \mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n)) - (\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\big(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n)) - \mathsf{R}_{\widetilde{\mathcal{F}}}(\varepsilon\boldsymbol{E}(t_n))\big) = \mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n)) - (\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathbb{E}_\varepsilon(\boldsymbol{E}(t_n)),$$

by the definition of the spatial residual $\mathbb{E}_\varepsilon$. Hence

$$\mathsf{r}^n = -\frac{\Delta t}{2}\mathsf{ROT}\big((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathbb{E}_\varepsilon(\boldsymbol{E}(t_n))\big) + \mathsf{ROT}\Big(\frac{\Delta t}{2}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n)) - \int_{J_n}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t))\,dt\Big).$$

(ii) Time part. Since $|J_n| = \Delta t/2$, the last bracket is $\int_{J_n}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n)-\boldsymbol{E}(t))\,dt$. For every t, $\boldsymbol{E}(t)\in\boldsymbol{V}_0$: its edge circulations are single valued and vanish on boundary edges, and the commuting property of Proposition 3.2 together with Faraday's law gives $\mathsf{ROT}(\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t))) = \mathsf{R}_{\mathcal{F}}(\nabla\times\boldsymbol{E}(t)) = -\mathsf{R}_{\mathcal{F}}(\partial_t\boldsymbol{B}(t))$, with $\boldsymbol{B}=\boldsymbol{\mu}\boldsymbol{H}$. The time integral commutes with the linear maps ROT and $\mathsf{R}_{\mathcal{F}}$, hence

$$\mathsf{ROT}\Big(\int_{J_n}\mathsf{R}_{\mathcal{E}}(\boldsymbol{E}(t_n)-\boldsymbol{E}(t))\,dt\Big) = \int_{J_n}\mathsf{R}_{\mathcal{F}}\big(\partial_t\boldsymbol{B}(t)-\partial_t\boldsymbol{B}(t_n)\big)\,dt = \boldsymbol{\tau}^n,$$

which is (5.31).

(iii) The bound. By the primal CFL bound (5.13a), $\Delta t\|\mathsf{ROT}(\mathsf{g})\|_{\mu^{-1}} \le 2(1-\alpha)\|\mathsf{g}\|_\varepsilon$ for all $\mathsf{g}\in\mathcal{E}_0$, applied to $\mathsf{g} = (\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathbb{E}_\varepsilon(\boldsymbol{E}(t_n))\in\mathcal{E}_0$, with $1-\alpha\le 1$ and the isometry $\|(\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\boldsymbol{\phi}\|_\varepsilon = \|\boldsymbol{\phi}\|_{\varepsilon^{-1}}$ of (5.3), $\frac{\Delta t}{2}\|\mathsf{ROT}((\mathrm{H}_\varepsilon^{\mathcal{E}\widetilde{\mathcal{F}}})^{-1}\mathbb{E}_\varepsilon(\boldsymbol{E}(t_n)))\|_{\mu^{-1}} \le \|\mathbb{E}_\varepsilon(\boldsymbol{E}(t_n))\|_{\varepsilon^{-1}}$, and the triangle inequality in (5.31) gives (5.32).

C.2. **Proof of Lemma 5.12.** Set $\boldsymbol{w} := \boldsymbol{v}-\pi_0\boldsymbol{v}$, so $\nabla\times\boldsymbol{w} = \nabla\times\boldsymbol{v}$ and $|\boldsymbol{w}|_{[H^1(\mathfrak{c})]^3} = |\boldsymbol{v}|_{[H^1(\mathfrak{c})]^3}$ (the curl and the gradient of a constant vanish). The curve γ is a union of edges of tetrahedra $\mathfrak{t}\subset\mathfrak{c}$ of the barycentric submesh (a primal edge is split at its midpoint, a dual-edge piece is a single edge), and σ is a union of faces of such tetrahedra; the number of pieces is bounded and $\operatorname{diam}(\mathfrak{t})\approx h_{\mathfrak{c}}$ by mesh regularity, so the bounds below are applied piece by piece and summed.

(i) Scaled Sobolev bounds. On the reference tetrahedron $\widehat{\mathfrak{t}}$, the Sobolev embedding $H^1(\widehat{\mathfrak{t}})\hookrightarrow L^4(\widehat{\mathfrak{t}})$ and the trace theorem $H^1(\widehat{\mathfrak{t}})\to L^4(\partial\widehat{\mathfrak{t}})$ hold . Scaling to $\mathfrak{t}$ (a factor $h_{\mathfrak{c}}^{3/4}$ for an L^4 volume norm, $h_{\mathfrak{c}}^{1/2}$ for an L^4 surface norm, $h_{\mathfrak{c}}^{3/2}$ for an L^2 volume norm, $h_{\mathfrak{c}}^{1/2}$ for an H^1 seminorm) and the Poincaré inequality for the mean-free field, $\|\boldsymbol{w}\|_{[L^2(\mathfrak{c})]^3}\lesssim h_{\mathfrak{c}}|\boldsymbol{w}|_{[H^1(\mathfrak{c})]^3}$ (valid with a constant depending only on the mesh regularity: by the assumptions of Section 3.1, $\mathfrak{c}$ is star-shaped with respect to a ball of radius $\approx h_{\mathfrak{c}}$, see [24, 44]), give

$$\|\boldsymbol{w}\|_{[L^4(\mathfrak{t})]^3}\lesssim h_{\mathfrak{c}}^{\frac14}|\boldsymbol{v}|_{[H^1(\mathfrak{c})]^3},\qquad \|\boldsymbol{w}\|_{[L^4(\partial\mathfrak{t})]^3}\lesssim |\boldsymbol{v}|_{[H^1(\mathfrak{c})]^3},\qquad \|\boldsymbol{w}\|_{[L^2(\sigma)]^3}\lesssim h_{\mathfrak{c}}^{\frac12}|\boldsymbol{v}|_{[H^1(\mathfrak{c})]^3}. \tag{C.1}$$

(ii) Edge integral. By (C.1), $\boldsymbol{w}\in[L^4(\mathfrak{t})]^3$ and $\boldsymbol{w}\times\boldsymbol{n}\in[L^4(\partial\mathfrak{t})]^3$; moreover, $\nabla\times\boldsymbol{w} = \nabla\times\boldsymbol{v}\in[L^4(\mathfrak{t})]^3$ by assumption. Thus, by the extended edge bound of Ern and Guermond [25, Prop. 17.9, eq. (17.25)] with $p=4$ (based on the lifting argument of Amrouche et al. [1]: the edge circulation is rewritten by Stokes' theorem as a volume integral of $\nabla\times\boldsymbol{w}$ against the gradient of a lifting plus a boundary integral of $\boldsymbol{w}\times\boldsymbol{n}$),

$$\Big|\int_\gamma \boldsymbol{w}\cdot\boldsymbol{\tau}\Big| \lesssim h_{\mathfrak{c}}\big(h_{\mathfrak{c}}^{-\frac34}\|\boldsymbol{w}\|_{[L^4(\mathfrak{t})]^3} + h_{\mathfrak{c}}^{\frac14}\|\nabla\times\boldsymbol{w}\|_{[L^4(\mathfrak{t})]^3} + h_{\mathfrak{c}}^{-\frac12}\|\boldsymbol{w}\times\boldsymbol{n}\|_{[L^4(\partial\mathfrak{t})]^3}\big).$$

Inserting (C.1) term by term: gives (5.35).

(iii) Face integral. By the Cauchy–Schwarz inequality on σ, $|\int_\sigma\boldsymbol{w}\cdot\boldsymbol{n}|\le|\sigma|^{\frac12}\|\boldsymbol{w}\|_{[L^2(\sigma)]^3}$, and $|\sigma|\lesssim h_{\mathfrak{c}}^2$; the last bound in (C.1) gives $|\sigma|^{1/2}\|\boldsymbol{w}\|_{[L^2(\sigma)]^3}\lesssim h_{\mathfrak{c}}\cdot h_{\mathfrak{c}}^{1/2}|\boldsymbol{v}|_{[H^1(\mathfrak{c})]^3}$, which is (5.34).

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